Cyber Security Internship – PicoCTF Submission

Submitted By: Md Yasir
Platform: PicoCTF
Total Challenges Solved: 40
Categories Covered: Web Exploitation, Cryptography, Forensics, General Skills

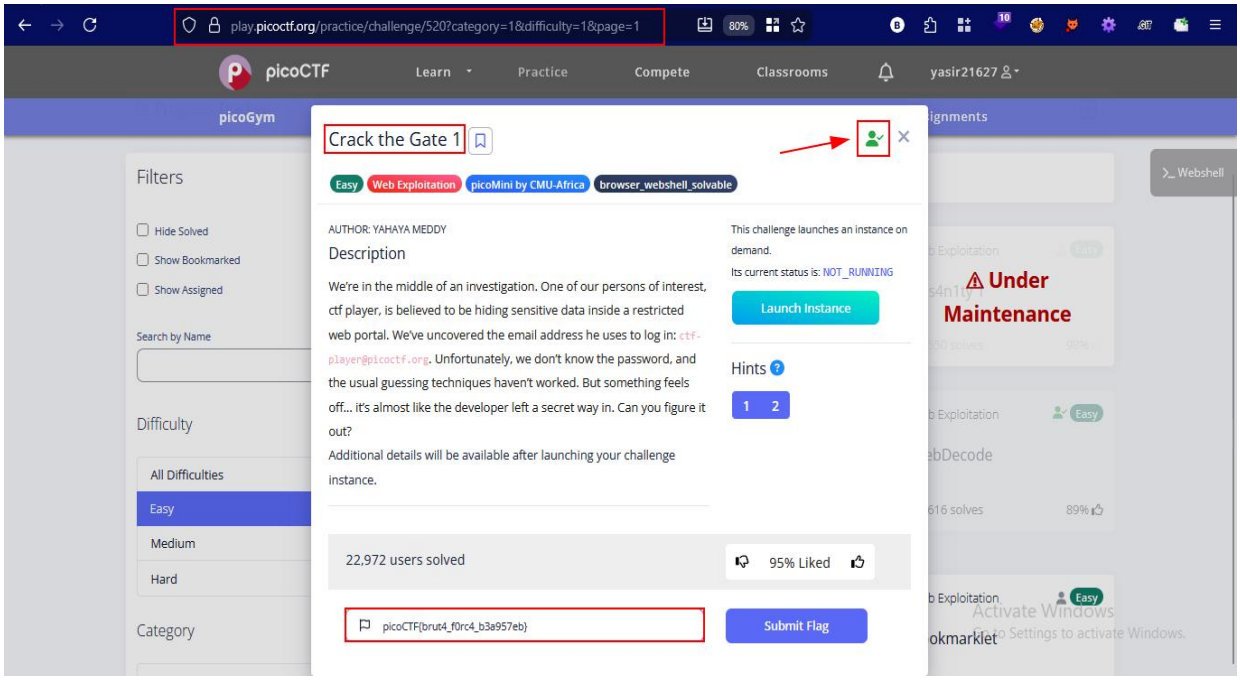
Introduction

Summary:
This report documents the successful completion of 40 PicoCTF challenges as part of the Cyber Security Internship submission. The completed challenges covered multiple domains including Web Exploitation, Cryptography, Forensics, and General Skills. Each challenge contributed to improving practical problem-solving ability, technical analysis, and cybersecurity understanding through hands-on learning.

1 Web Exploitation Challenges

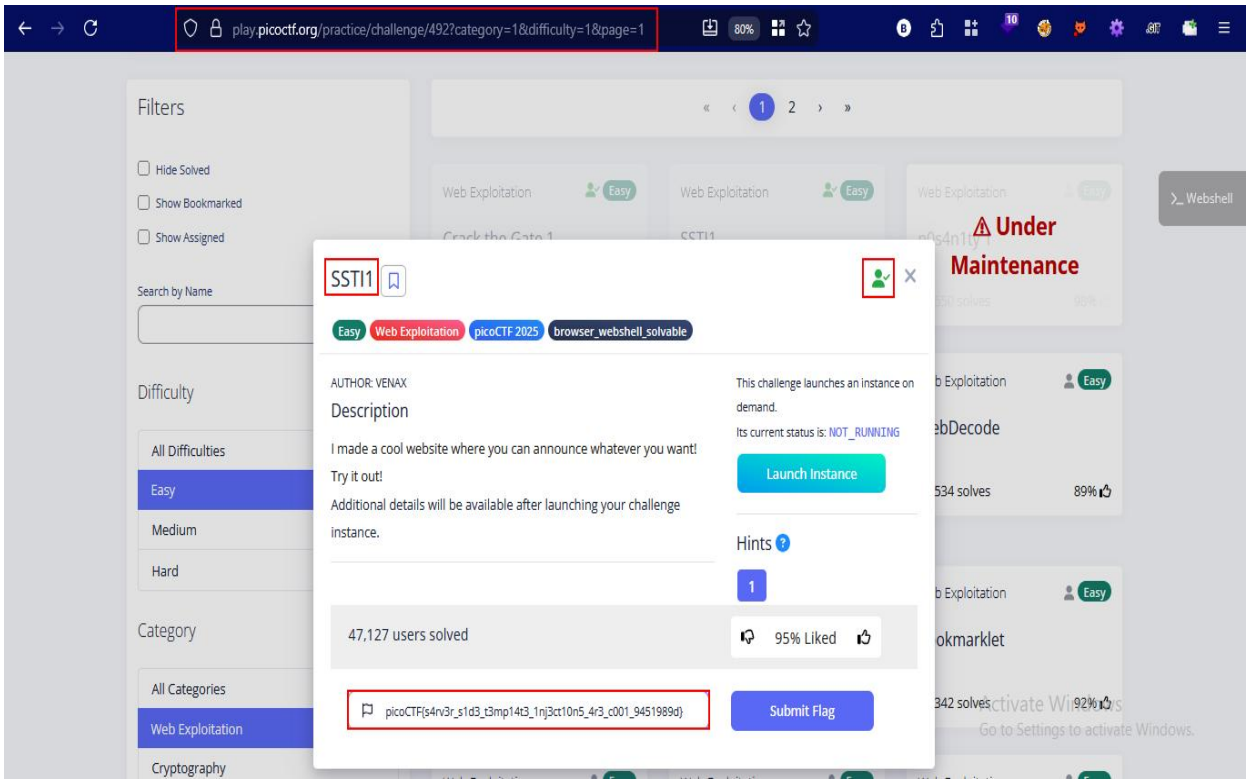
Easy Challenges

Task 1: Crack the Gate 1
Summary:
This challenge was completed successfully under the Web Exploitation category. It helped in understanding basic web challenge flow and controlled access logic.



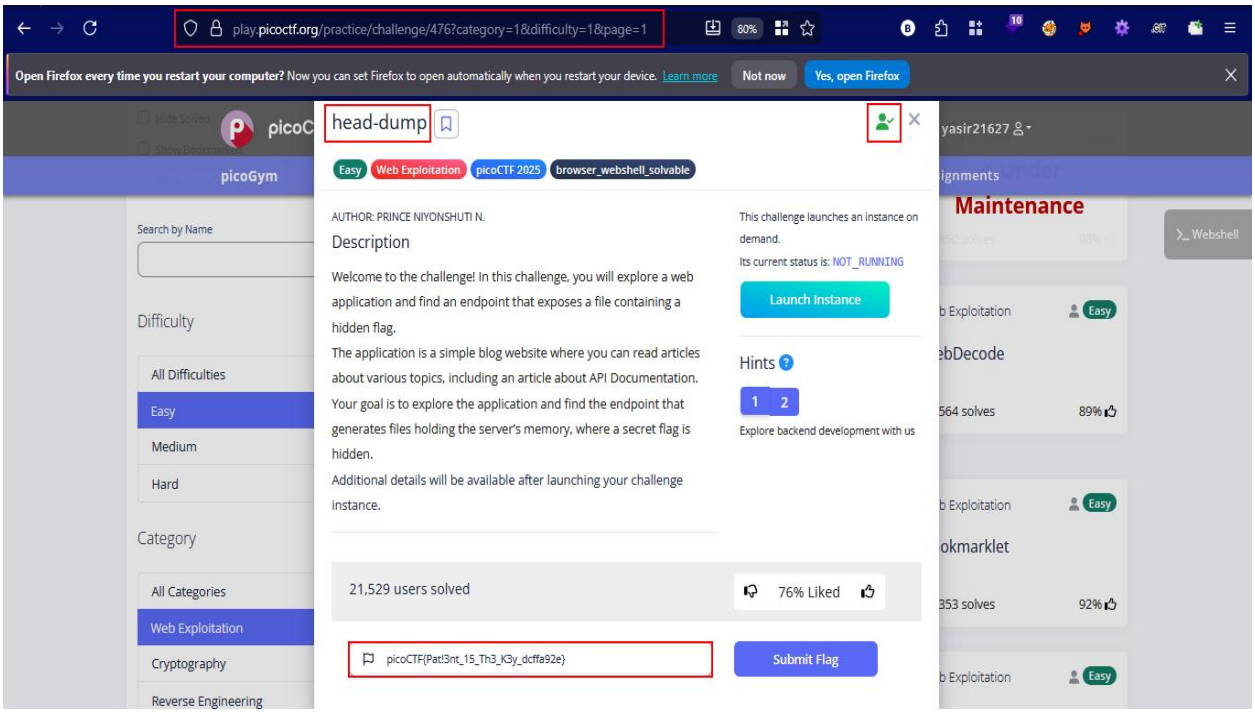
Task 2: SSTI1

Summary:
This challenge introduced server-side template injection concepts in a controlled environment. It improved understanding of unsafe input handling in template-based applications.



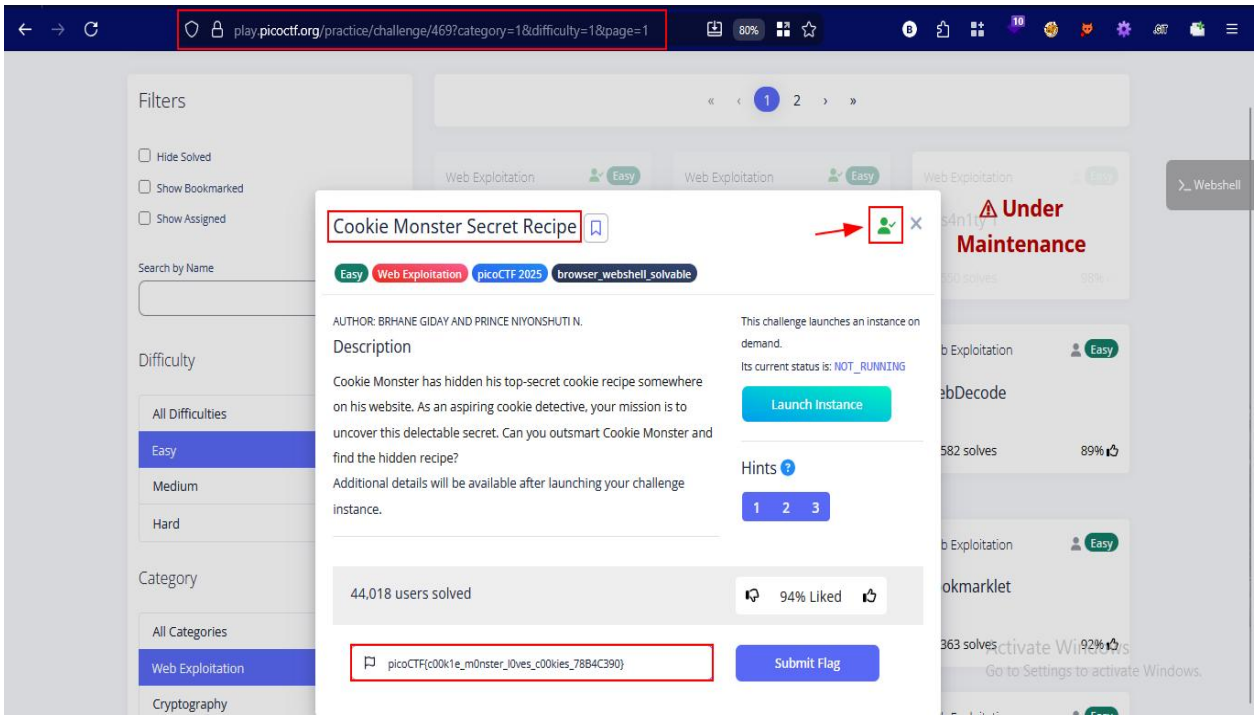
Task 3: head-dump

Summary:
This challenge focused on analysing web responses and exposed information. It strengthened practical observation and HTTP response analysis skills.



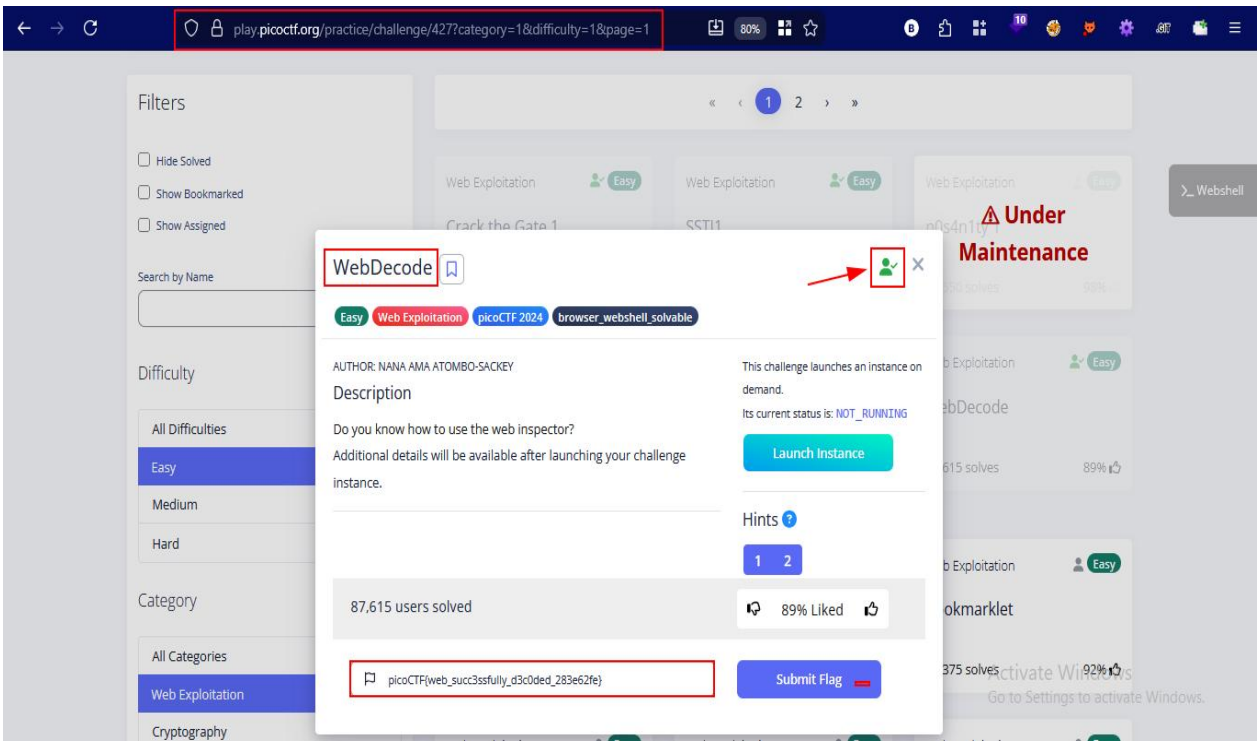
Task 4: Cookie Monster Secret Recipe

Summary:
This challenge involved reviewing client-side data and identifying exposed information. It improved practical debugging and web analysis skills.



Task 5: WebDecode

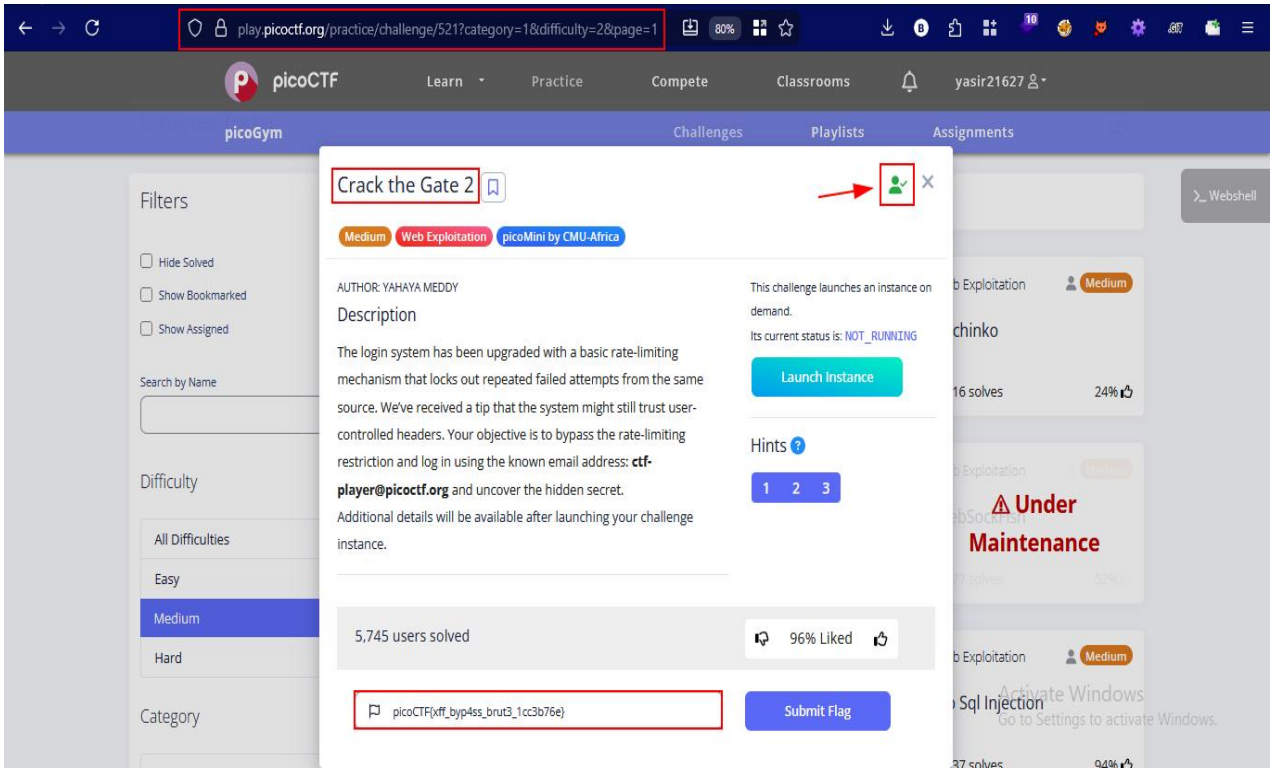
Summary:
This challenge focused on decoding web-related data to reveal hidden information.
It strengthened basic interpretation and analysis skills in a web context.



Medium Challenges

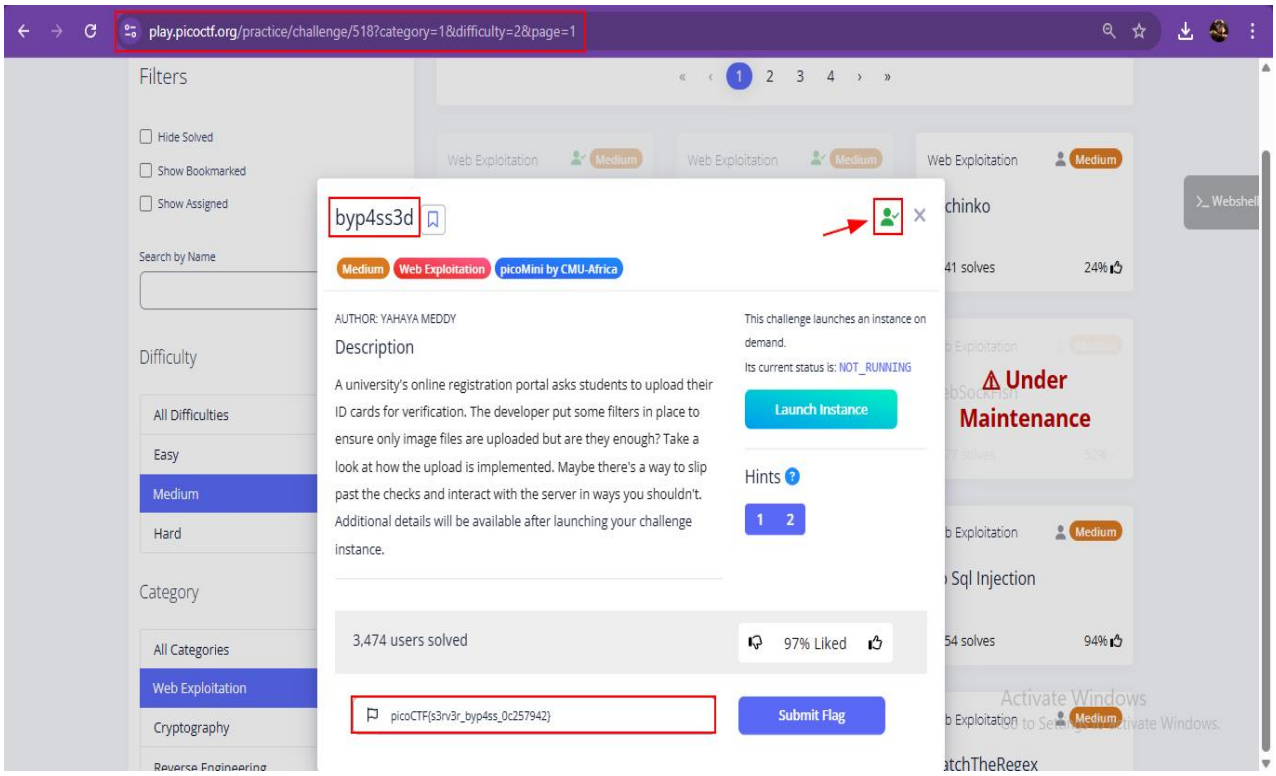
Task 1: Crack the Gate 2

Summary:
This challenge required deeper analysis of web application behavior and improved structured testing skills in a controlled environment.



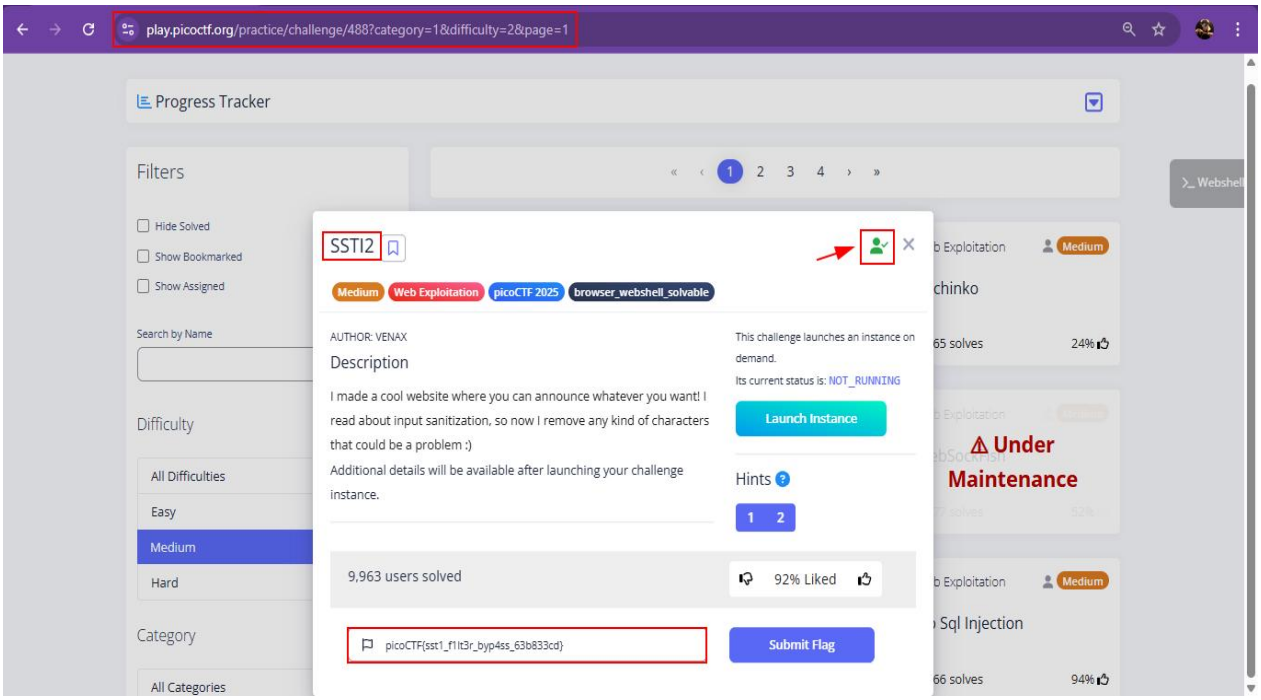
Task 2: byp4ss3d

Summary:
This challenge focused on bypass-oriented web logic and improved understanding of weak validation and insecure application checks.



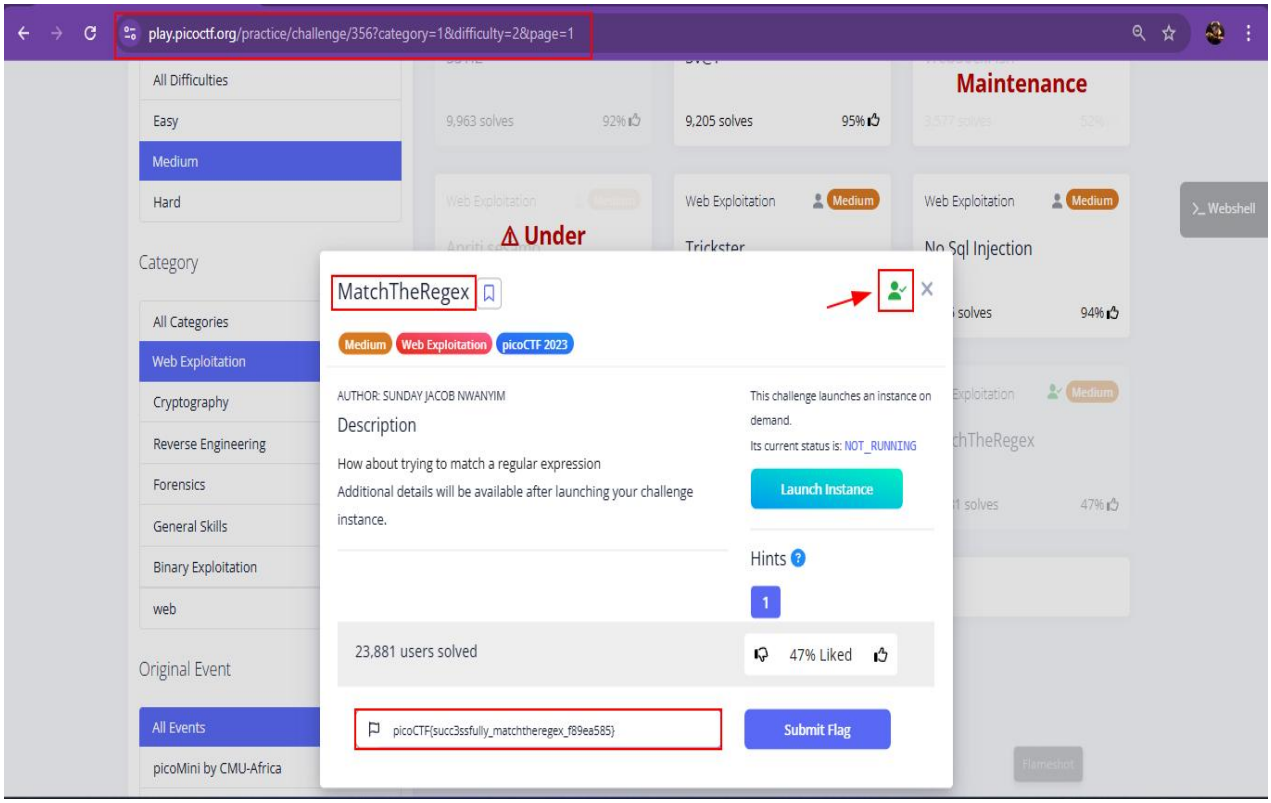
Task 3: SSTI2

Summary:
This challenge further explored server-side template injection concepts and strengthened understanding of insecure server-side rendering behavior.



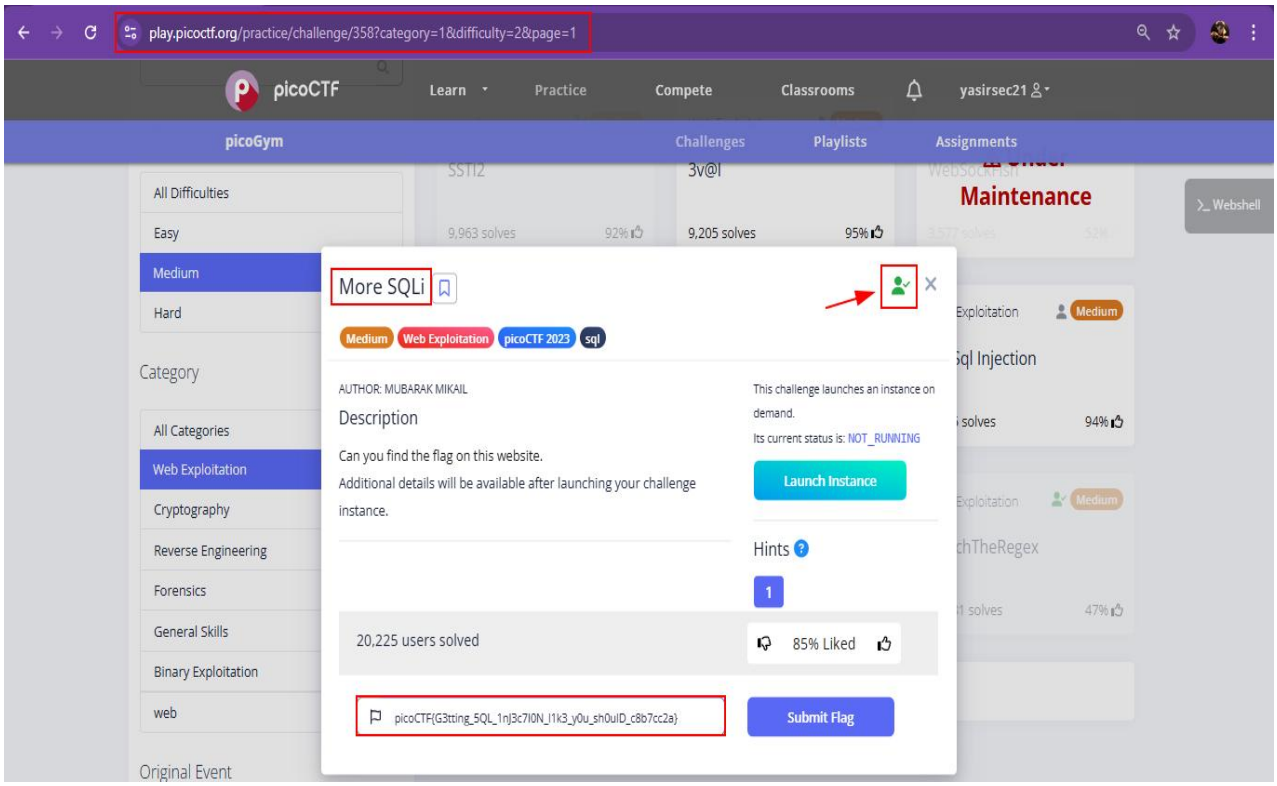
Task 4: MatchTheRegex

Summary:
This challenge focused on pattern matching and input handling logic in web applications. It improved analytical thinking and validation testing skills.



Task 5: More SQLi

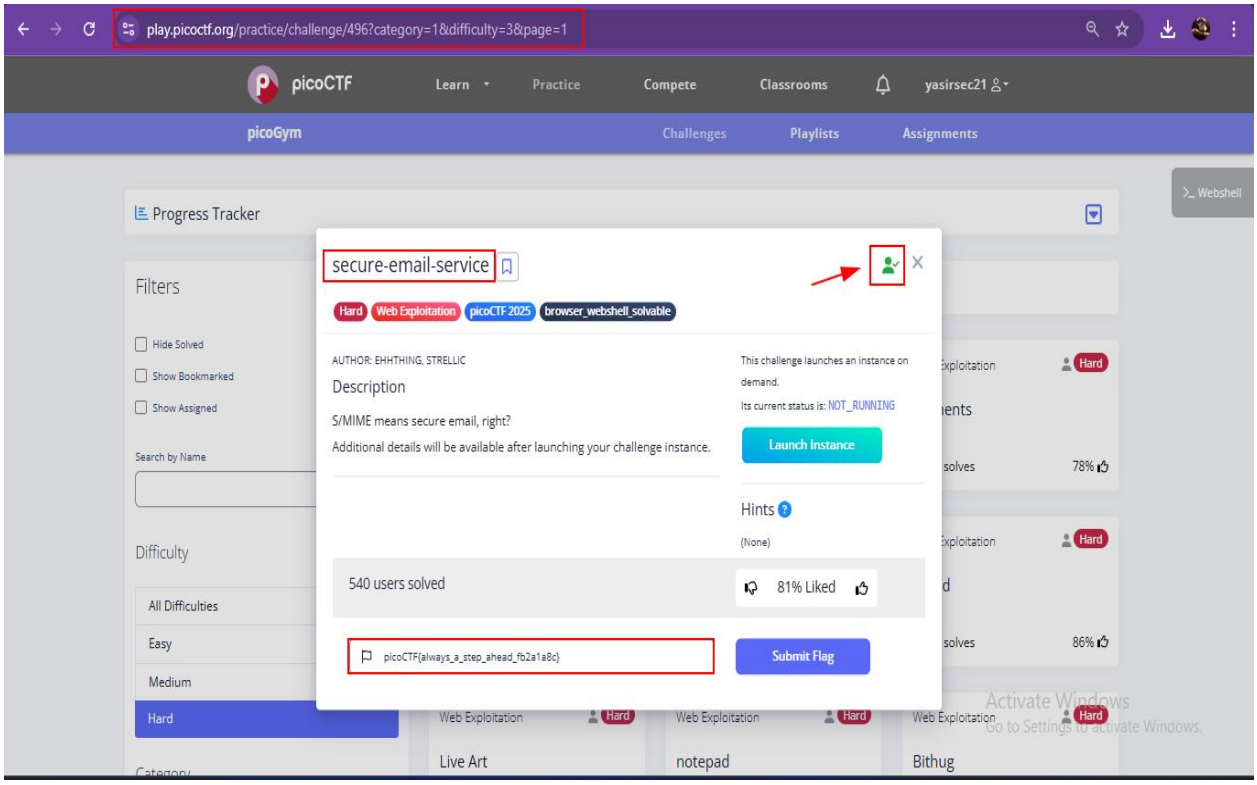
Summary:
This challenge provided practical exposure to advanced SQL injection concepts in a legal lab setup. It improved database-focused testing and exploitation understanding.



Hard Challenges

Task 1: secure-email-service

Summary:
This hard challenge required deeper web security analysis and structured exploitation logic. It strengthened practical problem-solving and advanced testing skills.



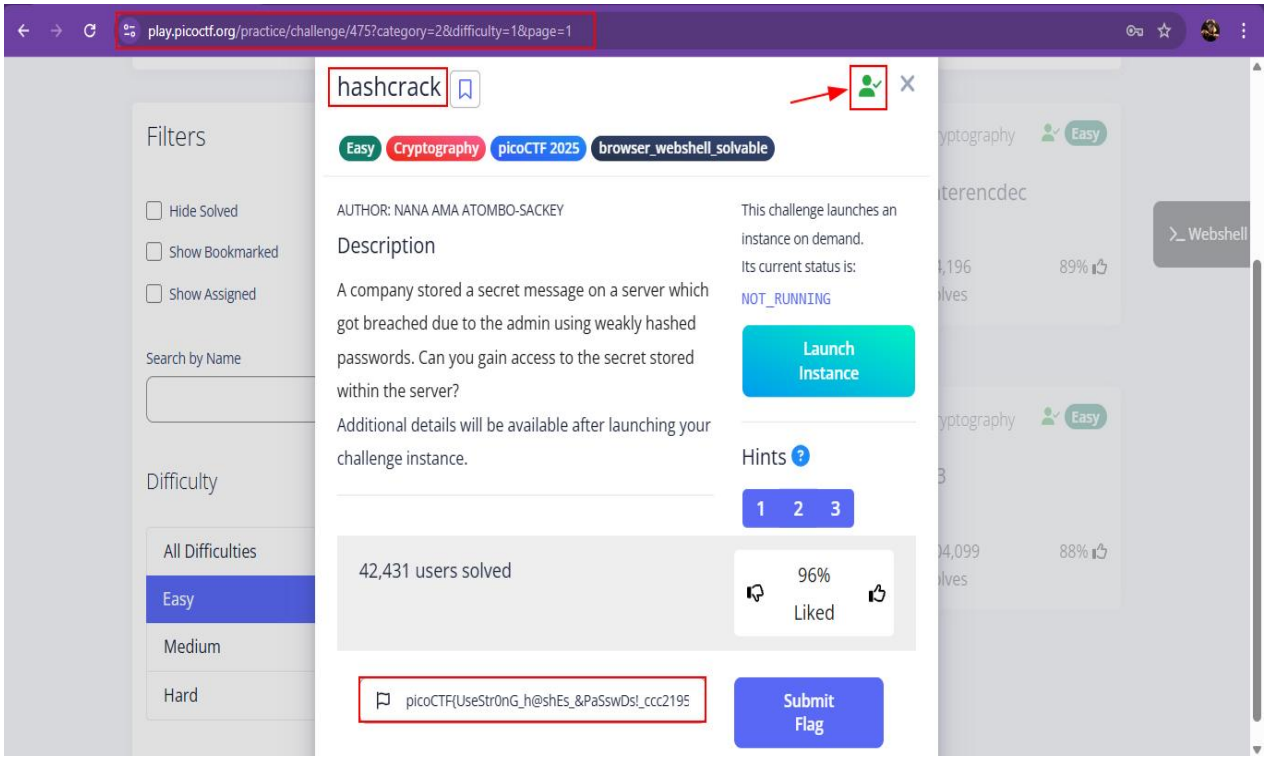
2 Cryptography Challenges

Easy Challenges

Task 1: hashcrack

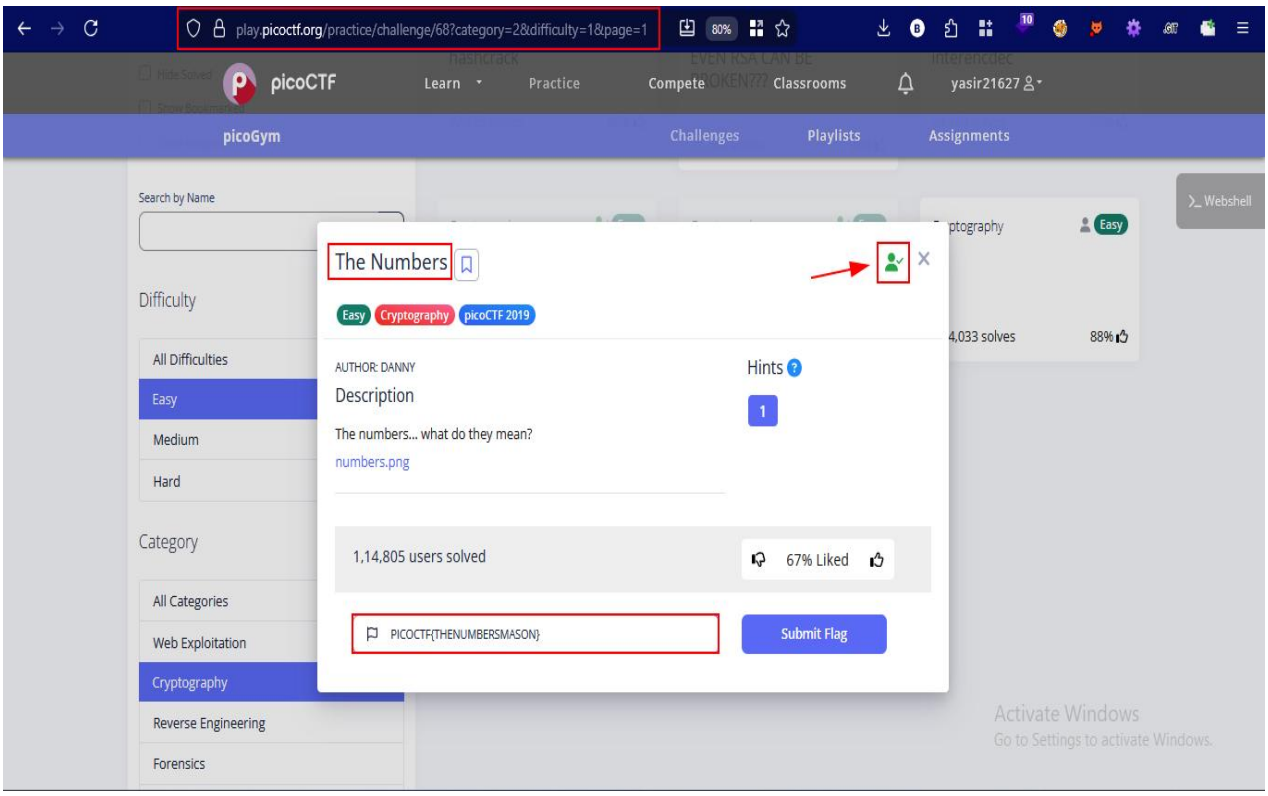
Summary:

This challenge focused on hash analysis in a controlled environment. It improved practical understanding of common cryptographic weaknesses.



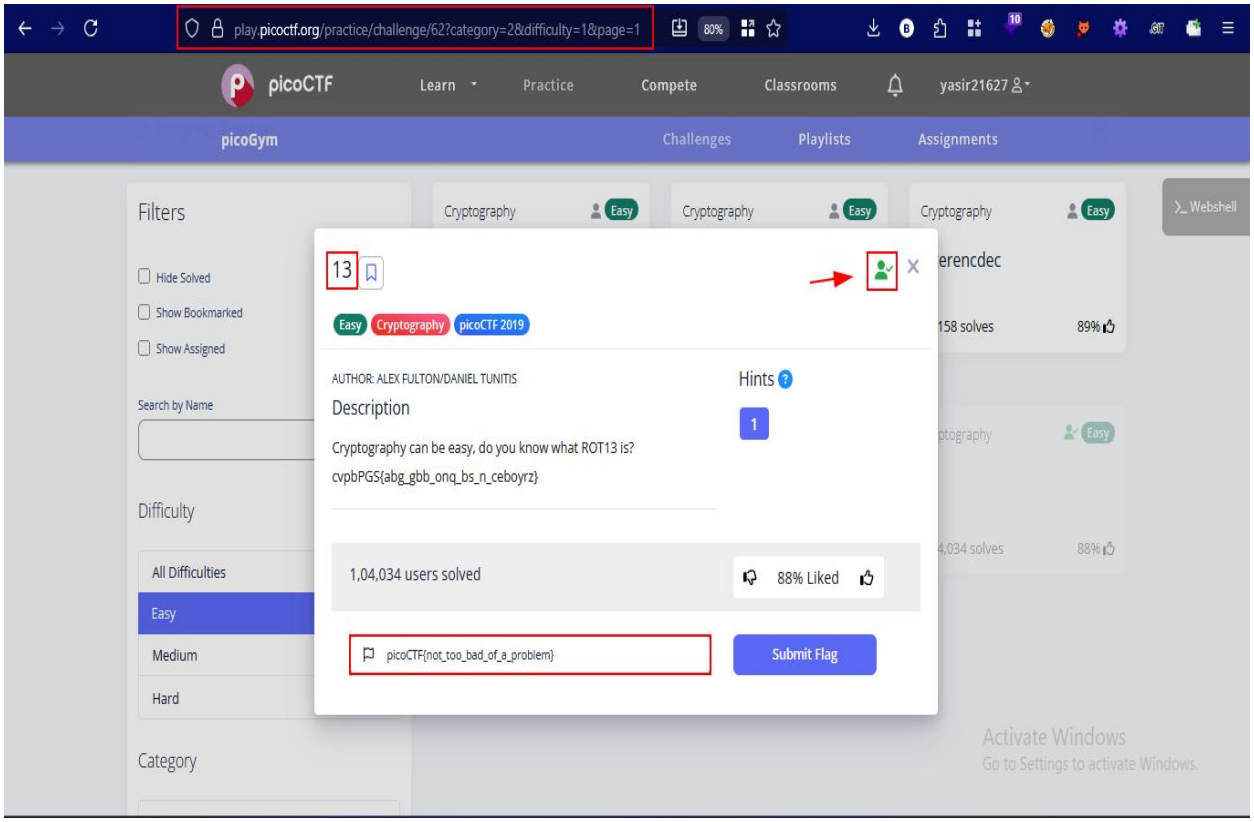
Task 2: The Numbers

Summary:
This challenge involved decoding numerical patterns to recover meaningful information. It strengthened logical interpretation and basic cryptographic reasoning.



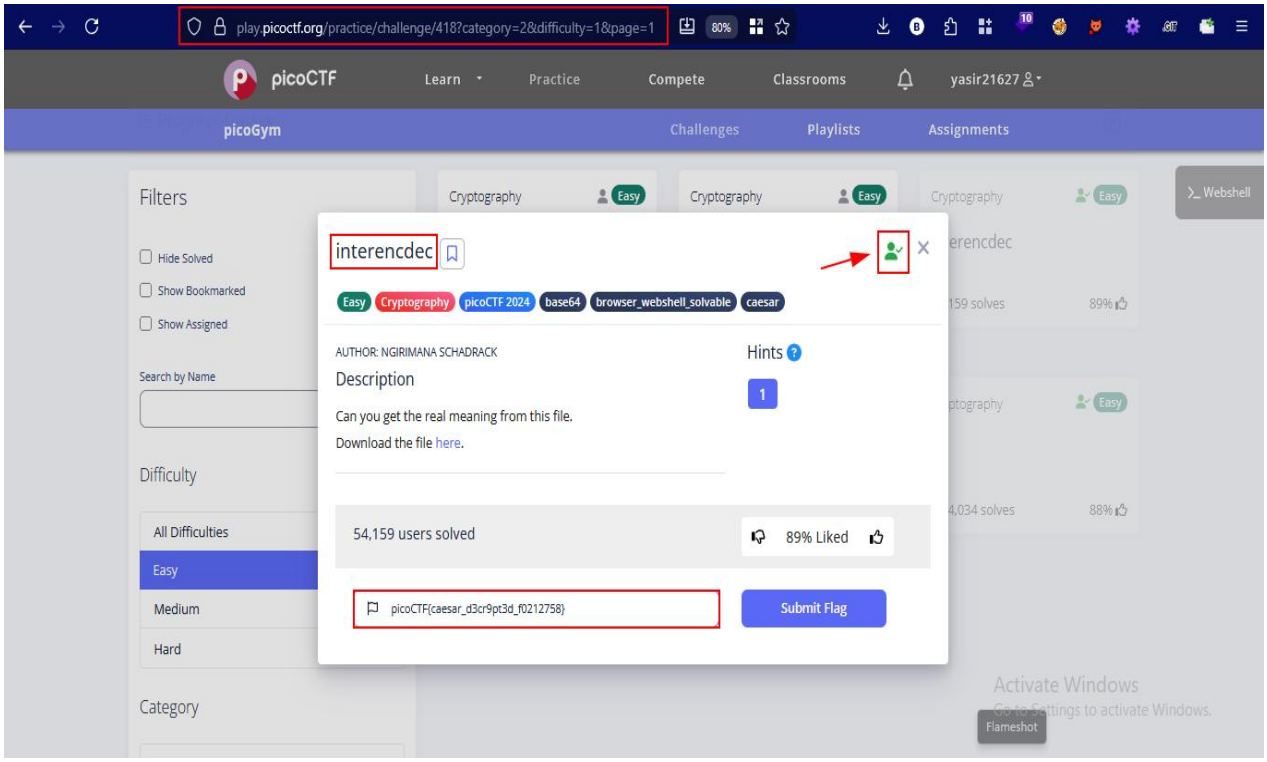
Task 3: 13

Summary:
This challenge focused on simple substitution-based transformation techniques. It improved understanding of introductory cryptographic methods.



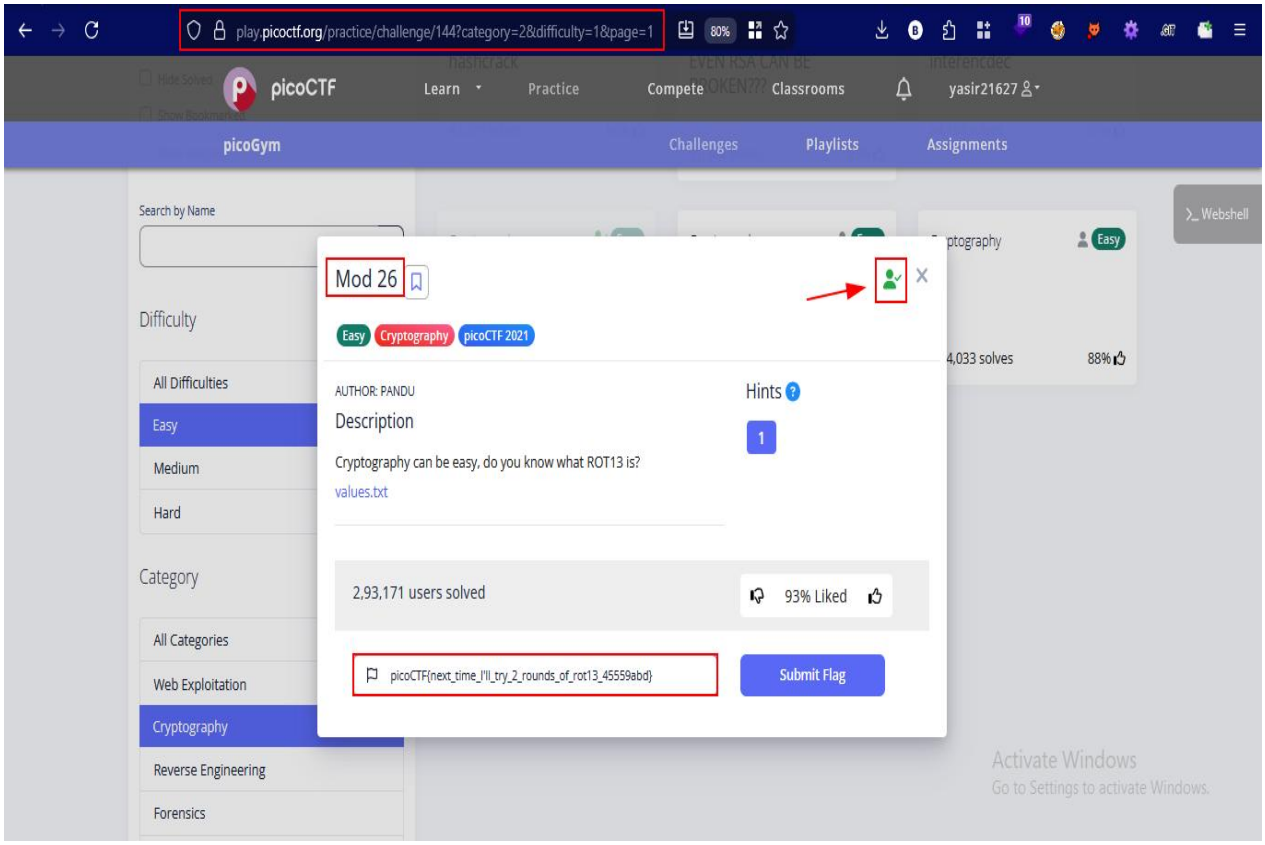
Task 4: interencdec

Summary:
This challenge involved mixed encoding and decoding analysis. It improved practical ability to recognise layered transformations in cryptographic data.



Task 5: Mod 26

Summary:
This challenge strengthened foundational knowledge of modular arithmetic and basic cipher transformation concepts.

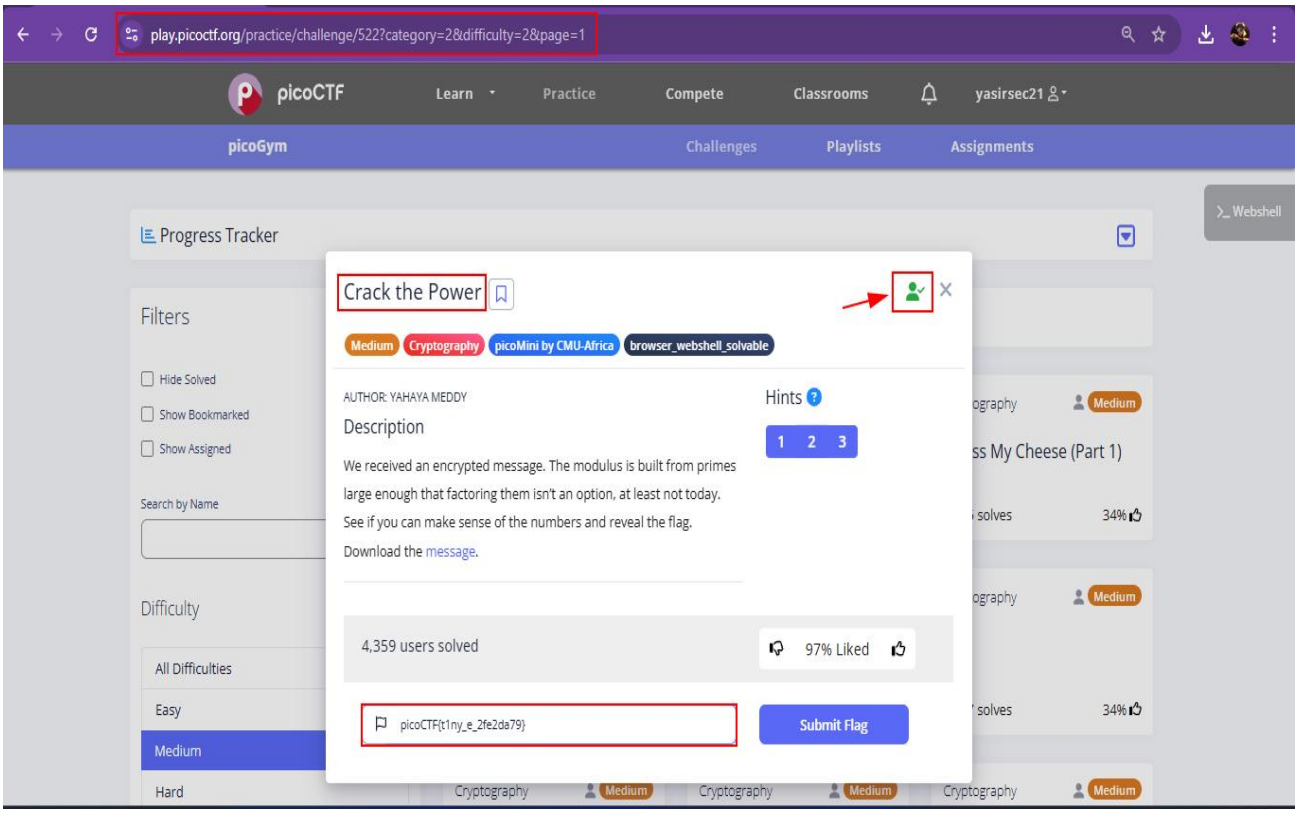


Medium Challenges

Task 1: Crack the Power

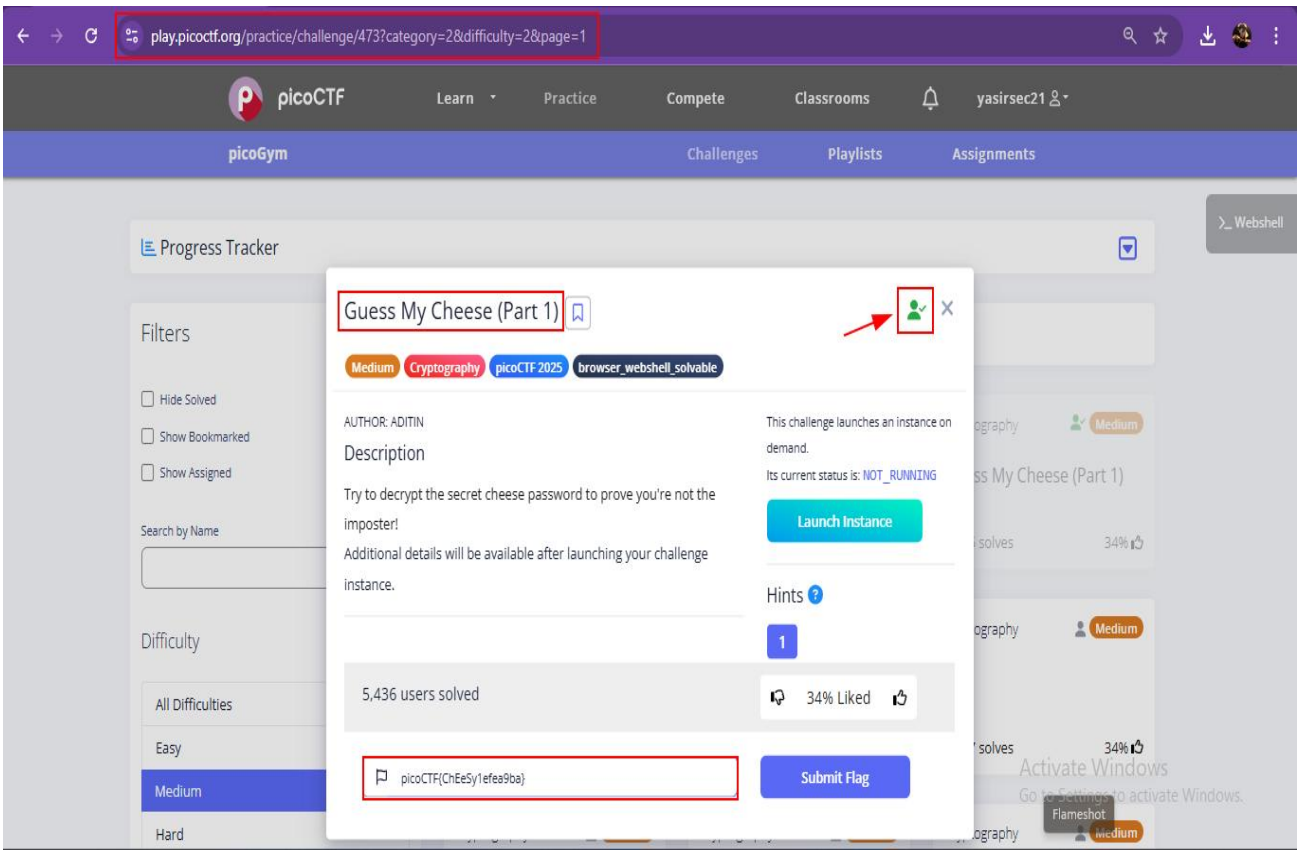
Summary:

This challenge required deeper cryptographic reasoning and improved practical analysis of protected data.



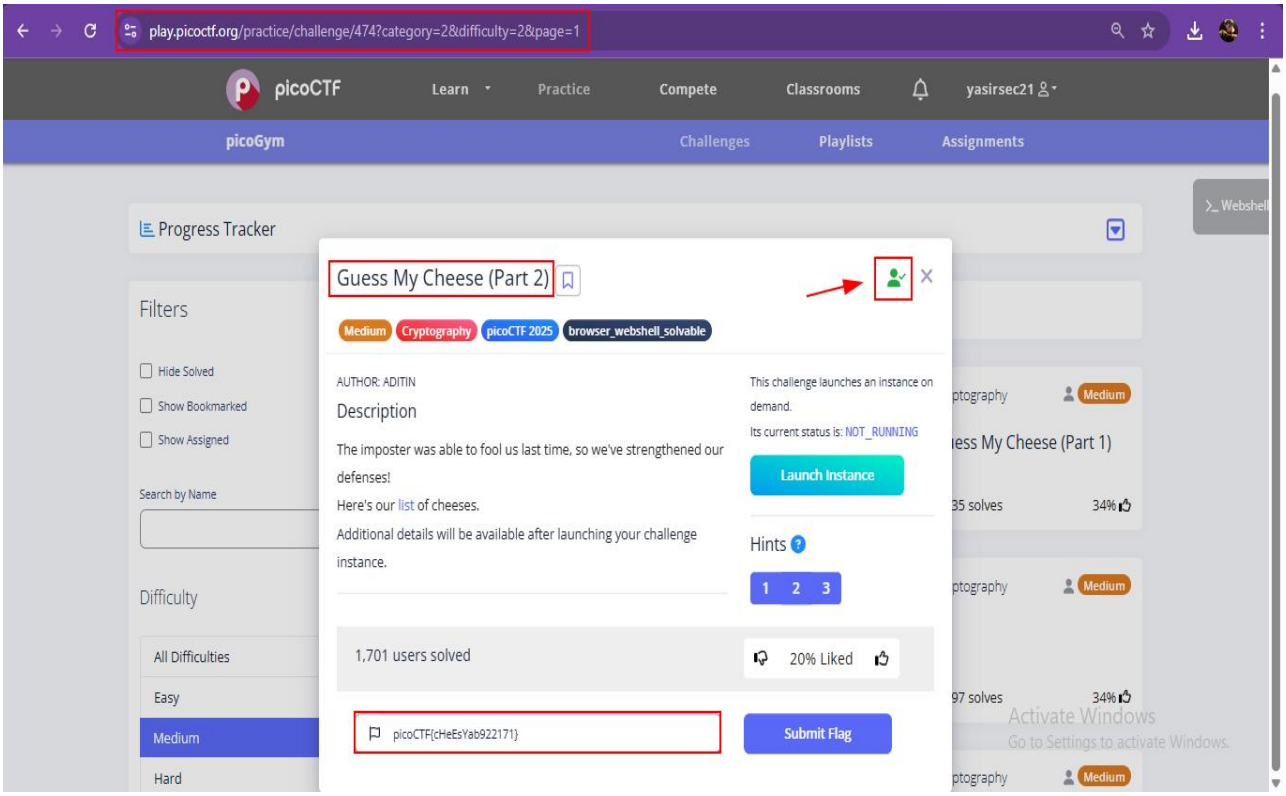
Task 2: Guess My Cheese (Part 1)

Summary:
This challenge introduced controlled cryptographic reasoning and improved understanding of challenge-based data interpretation.



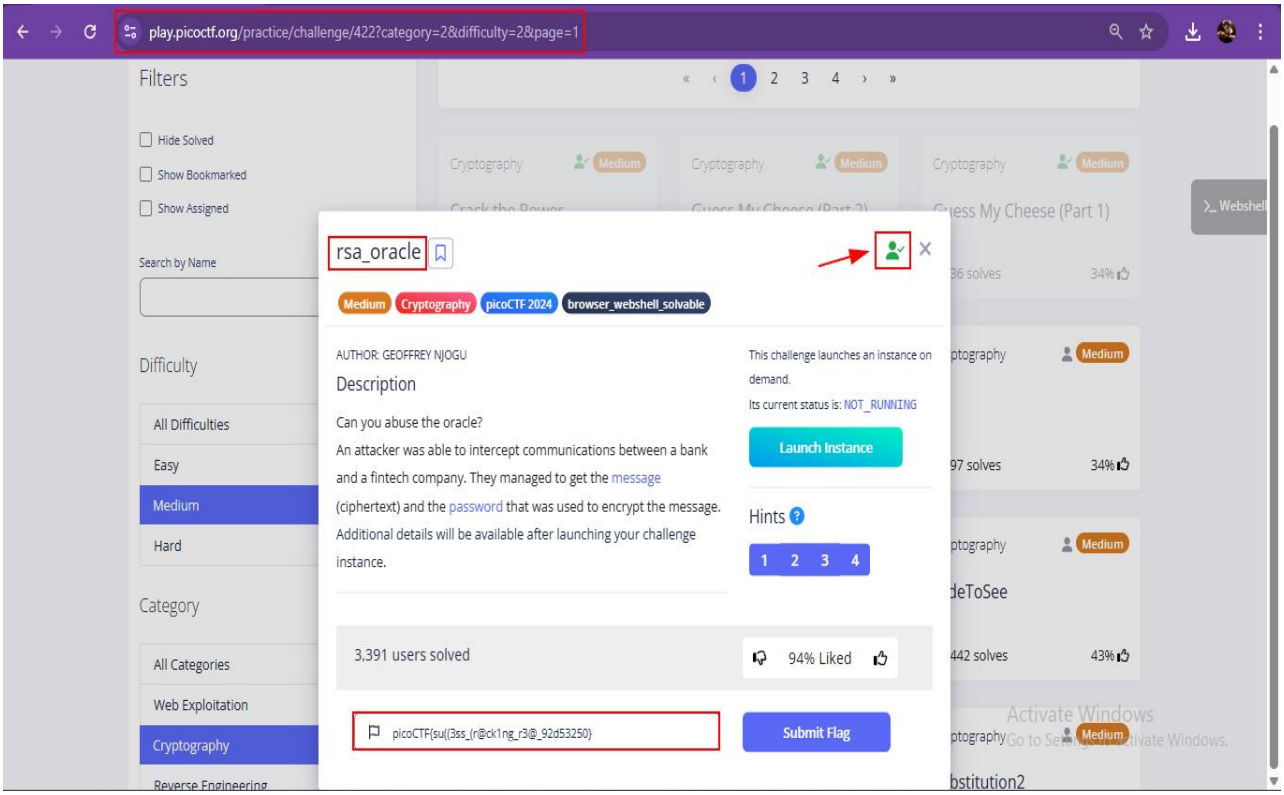
Task 3: Guess My Cheese (Part 2)

Summary:
This challenge focused on pattern-based cryptographic analysis and strengthened logical problem-solving skills.



Task 4: rsa_oracle

Summary:
This challenge focused on RSA-related concepts and improved practical understanding of oracle-based cryptographic weaknesses.

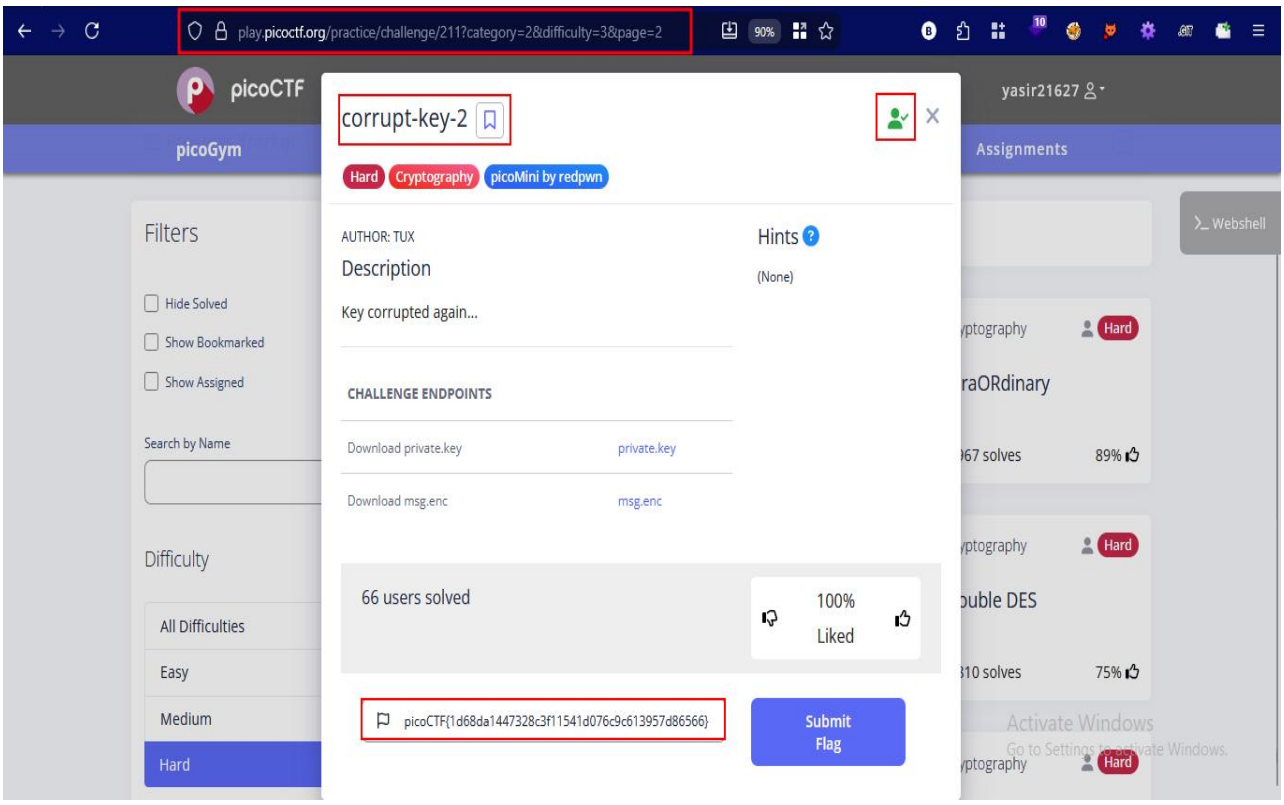


Hard Challenges

Task 1: Corrupt Key 2

Summary:

This hard challenge required advanced cryptographic analysis and structured reasoning. It strengthened confidence in handling complex crypto problems.



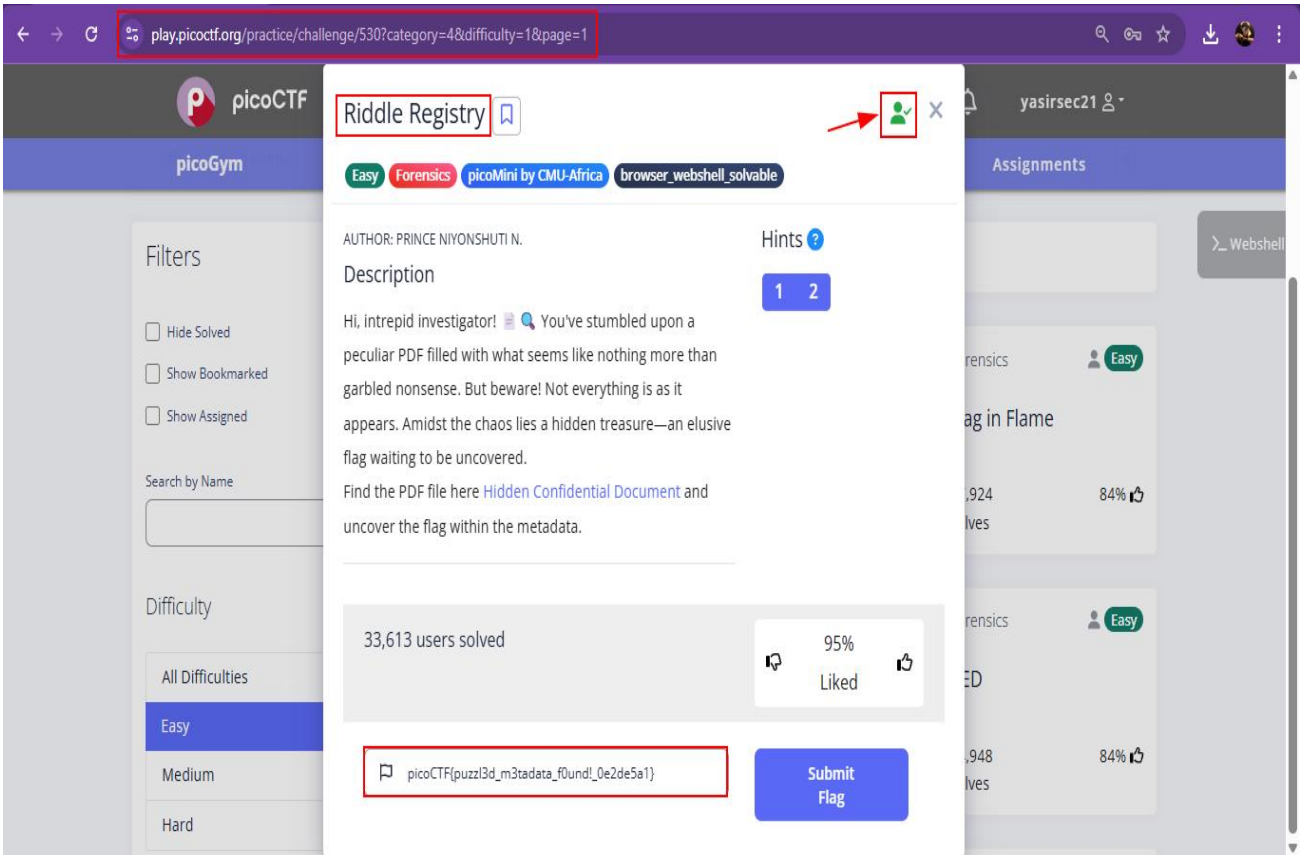
3 Forensics Challenges

Easy Challenges

Task 1: Riddle Registry

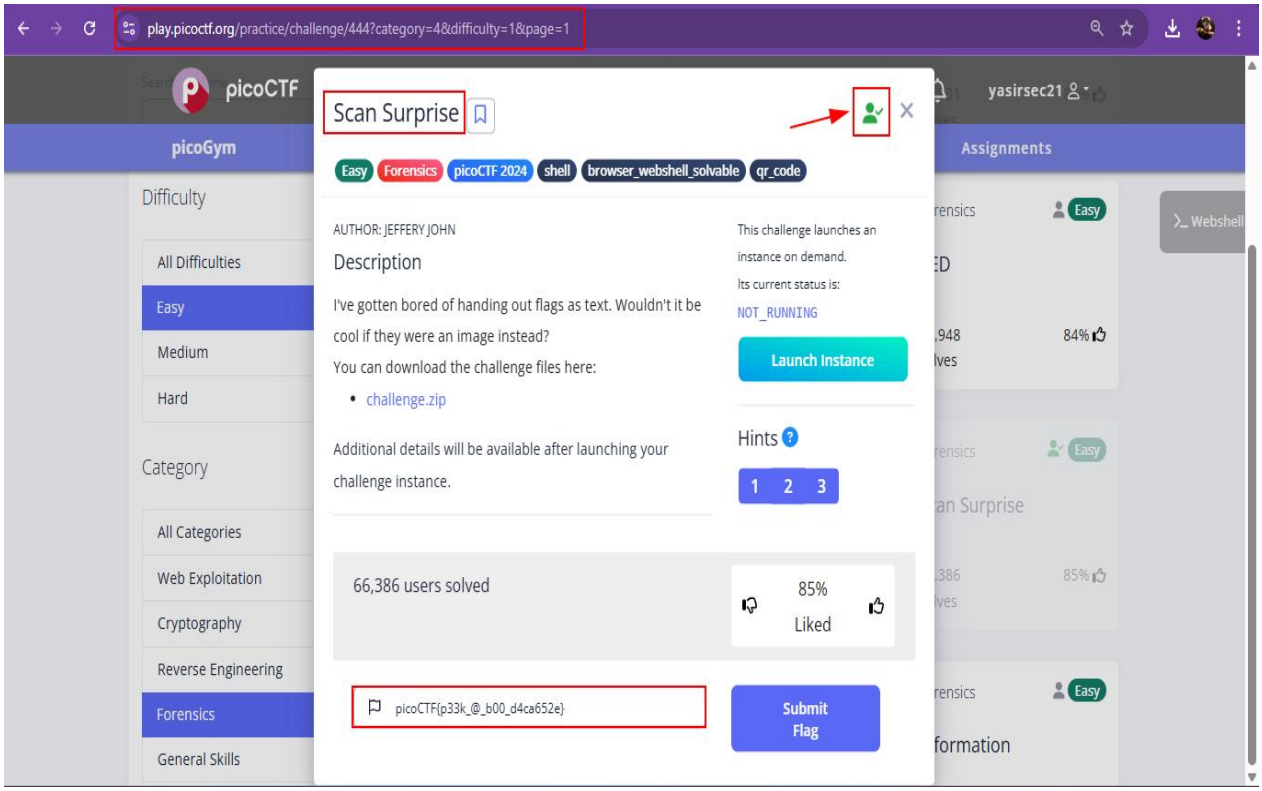
Summary:

This challenge focused on registry-based forensic analysis and improved practical investigation skills.



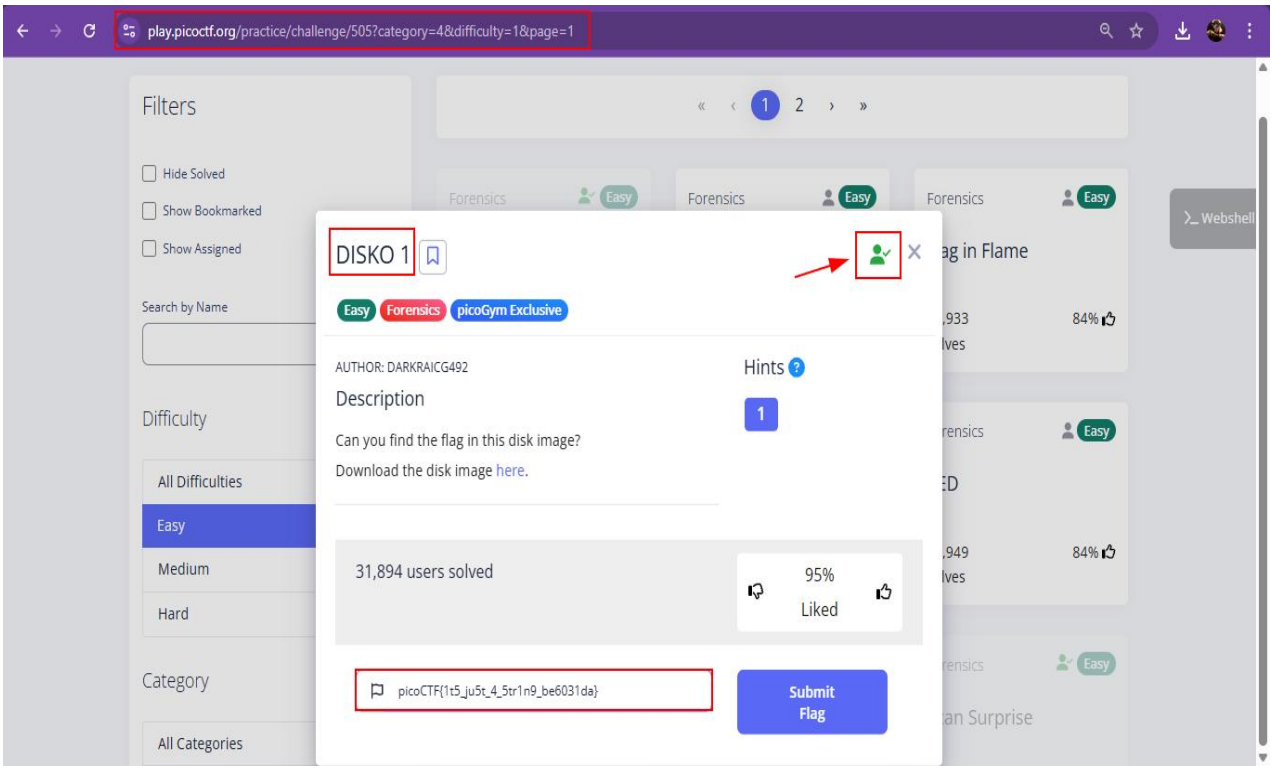
Task 2: Scan Surprise

Summary:
This challenge involved analysing provided forensic artefacts to identify hidden or relevant information. It strengthened digital evidence review skills.



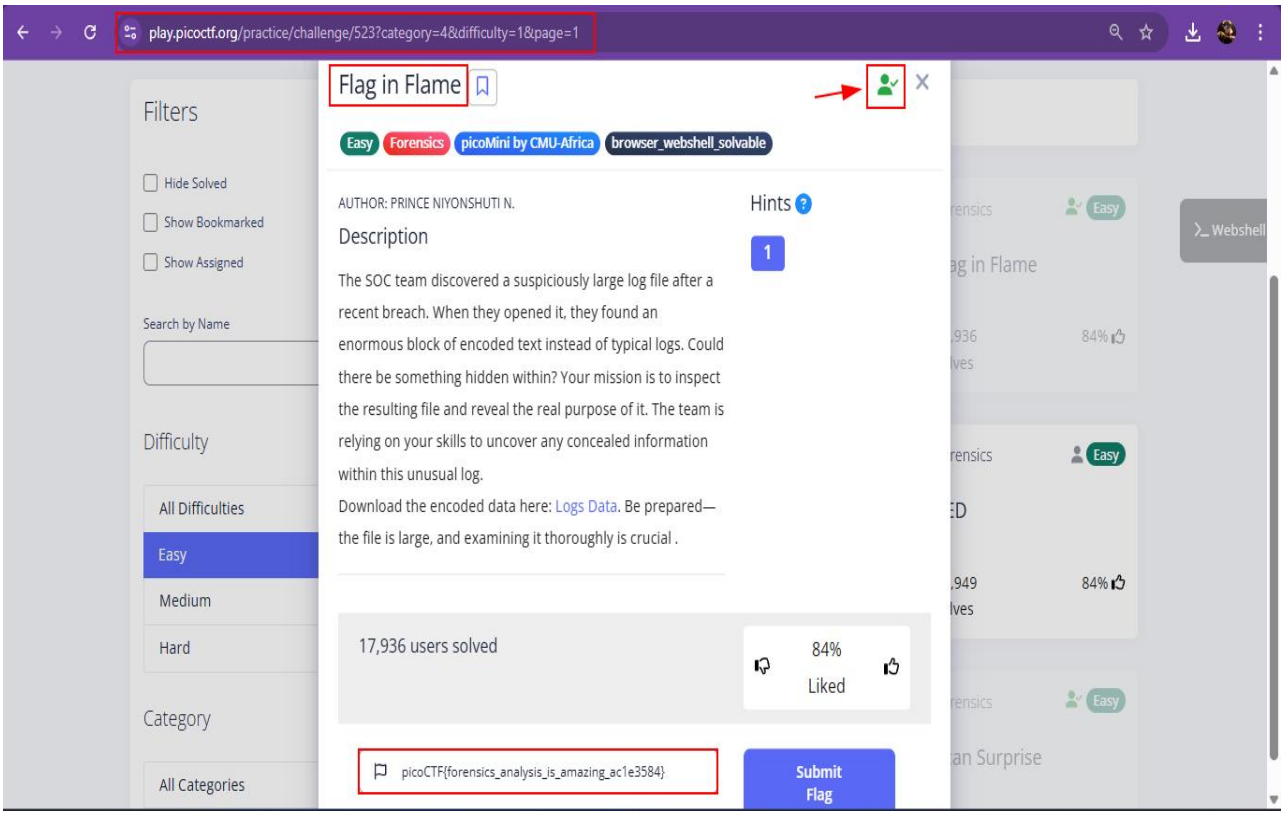
Task 3: DISKO 1

Summary:
This challenge focused on disk-related artefact inspection and improved foundational forensic analysis ability.



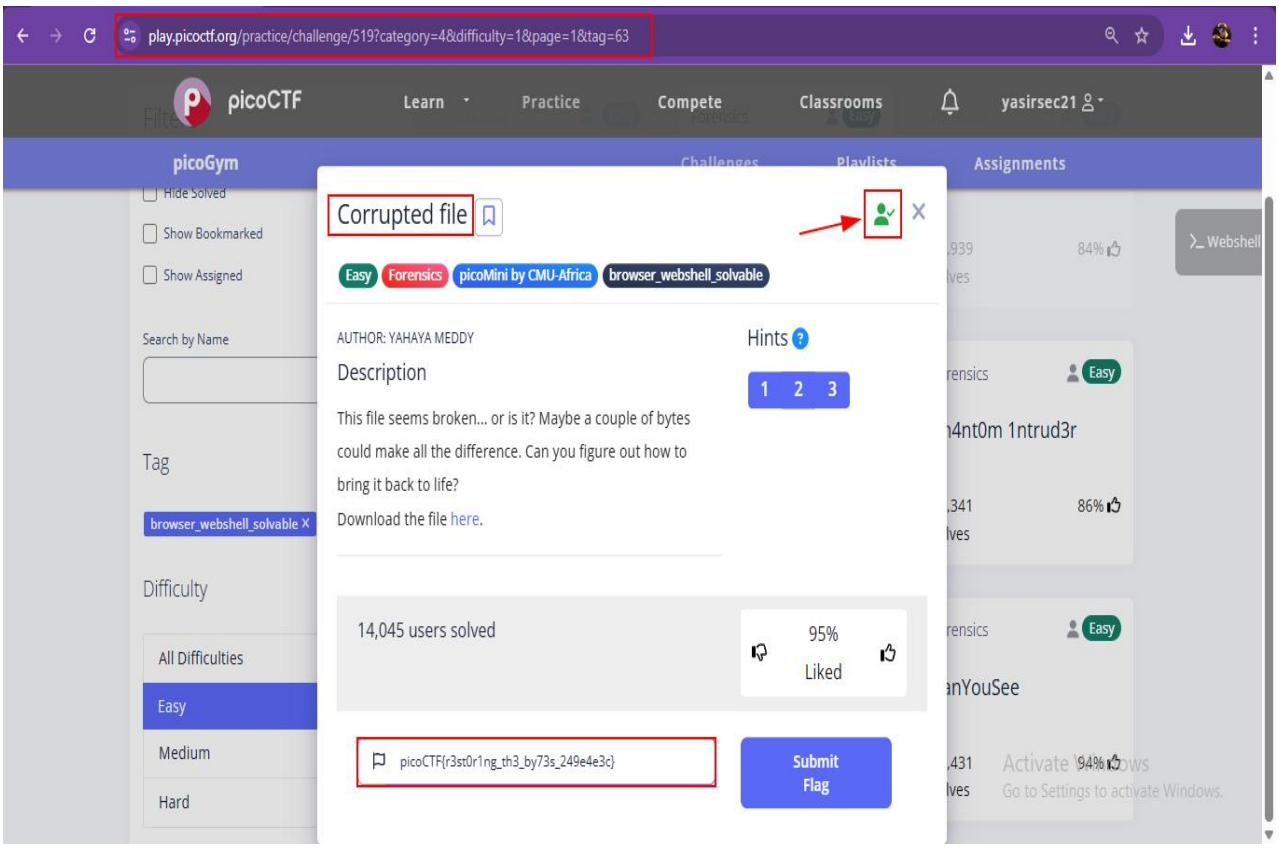
Task 4: Flag in Flame

Summary:
This challenge required recovering important information from forensic evidence. It strengthened file inspection and evidence discovery skills.



Task 5: Corrupted file

Summary:
This challenge focused on analysing damaged or altered file artefacts. It improved practical understanding of file recovery and inspection.

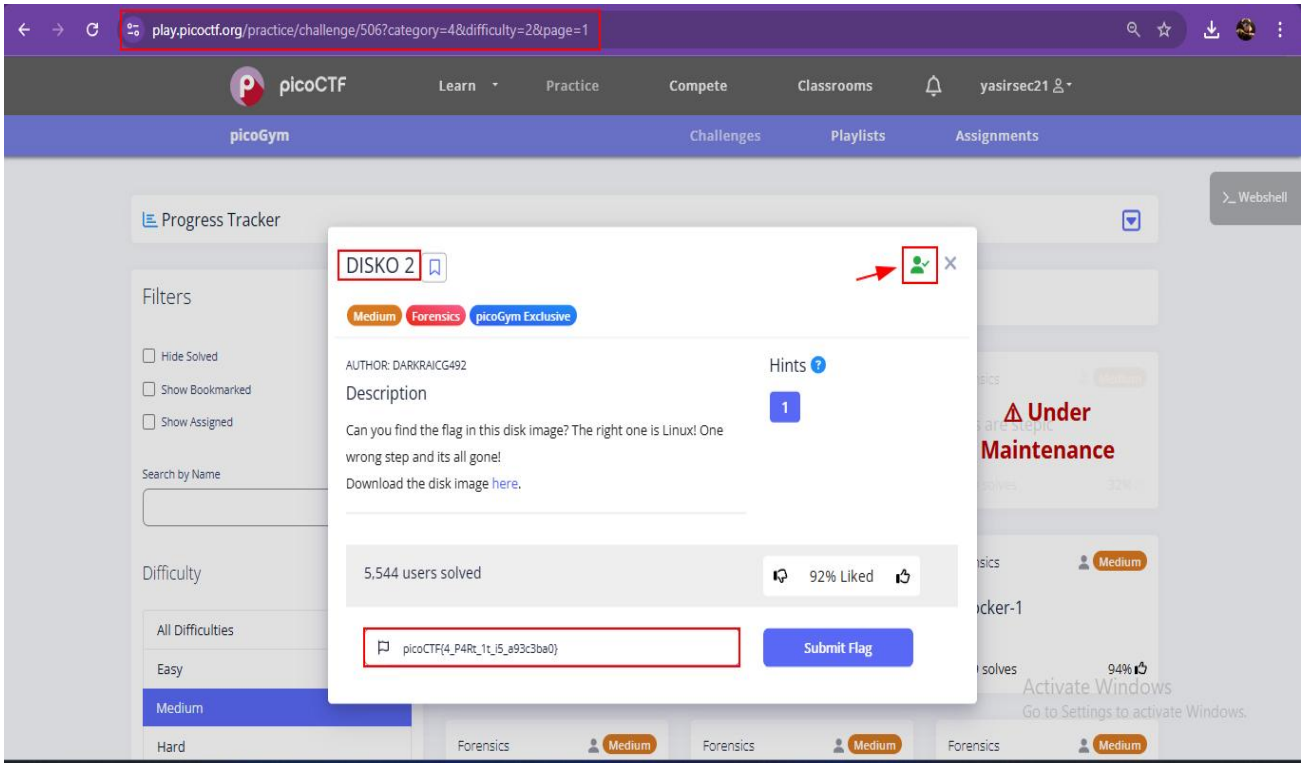


Medium Challenges

Task 1: DISKO 2

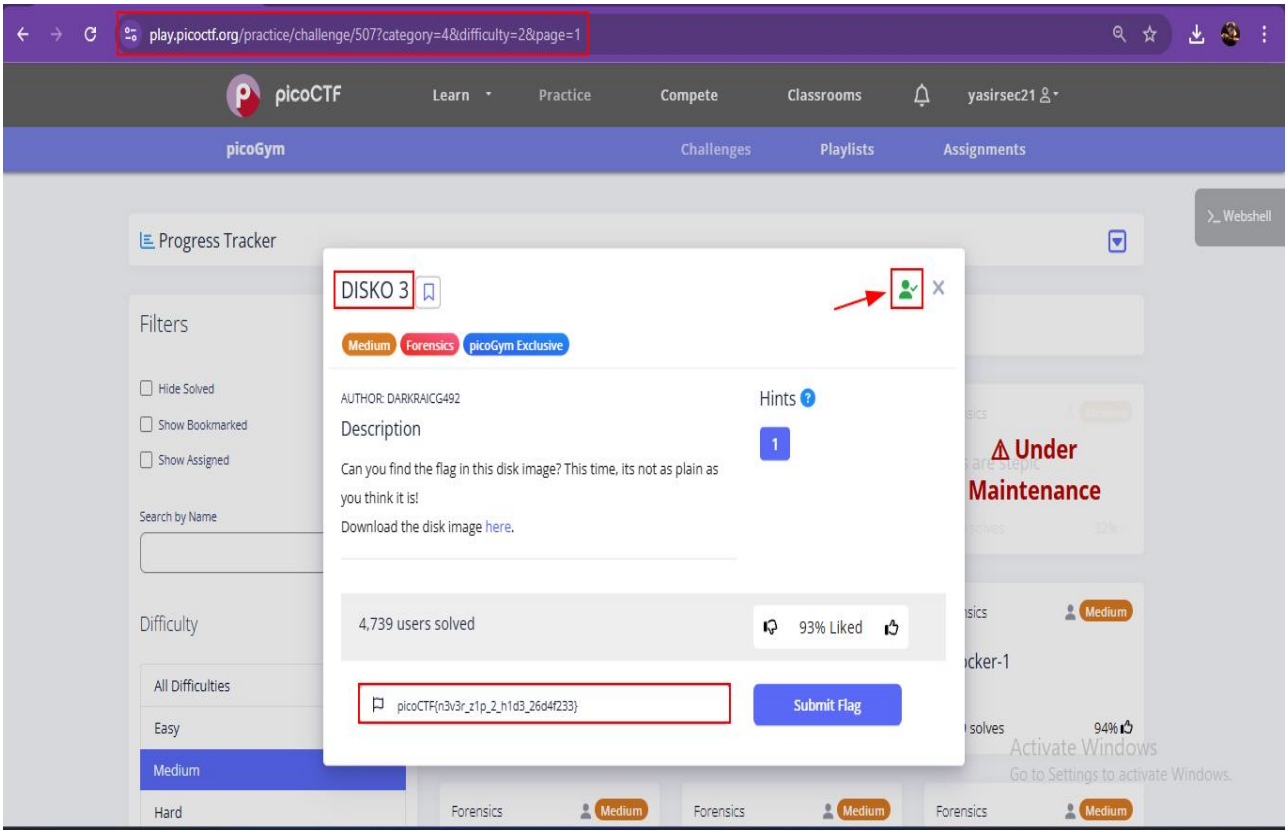
Summary:

This challenge strengthened practical understanding of disk-based evidence analysis in a controlled environment.



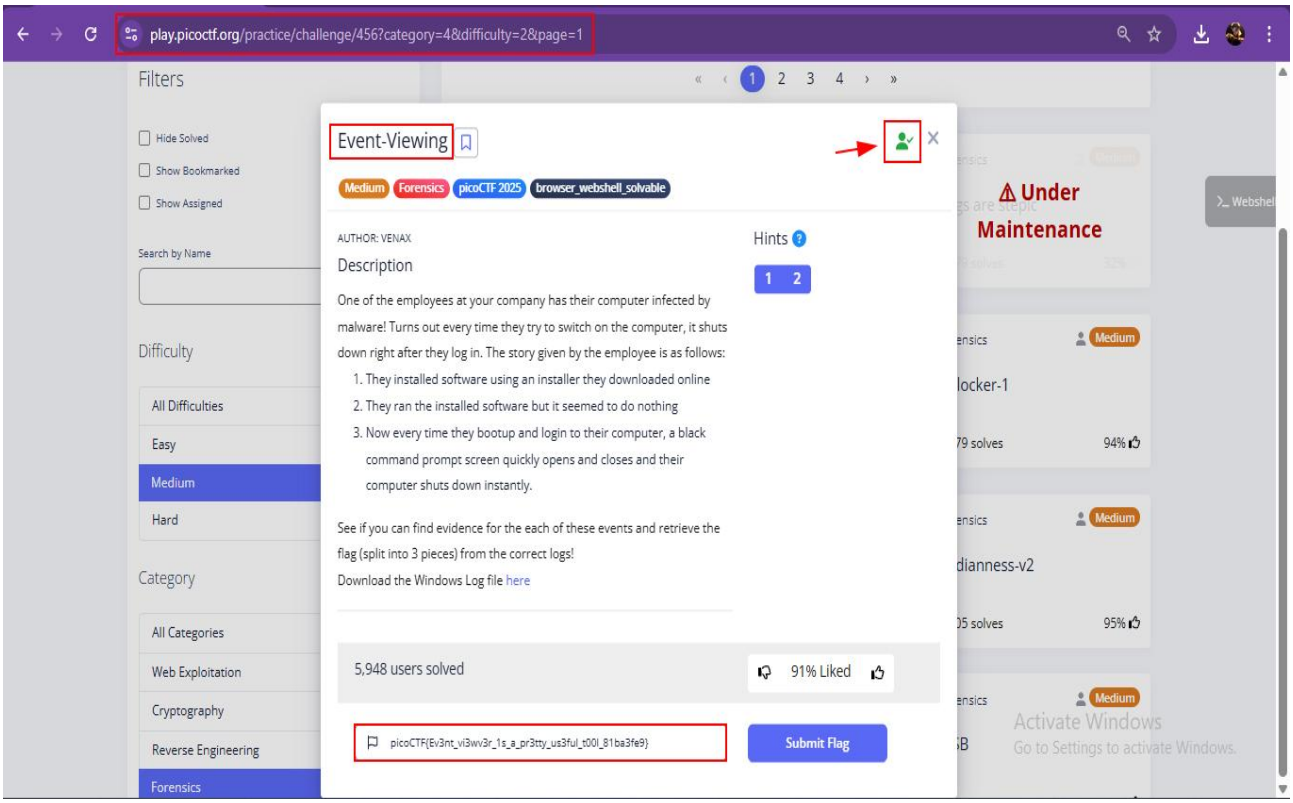
Task 2: DISKO 3

Summary:
This challenge involved deeper inspection of disk artefacts and improved structured forensic investigation skills.



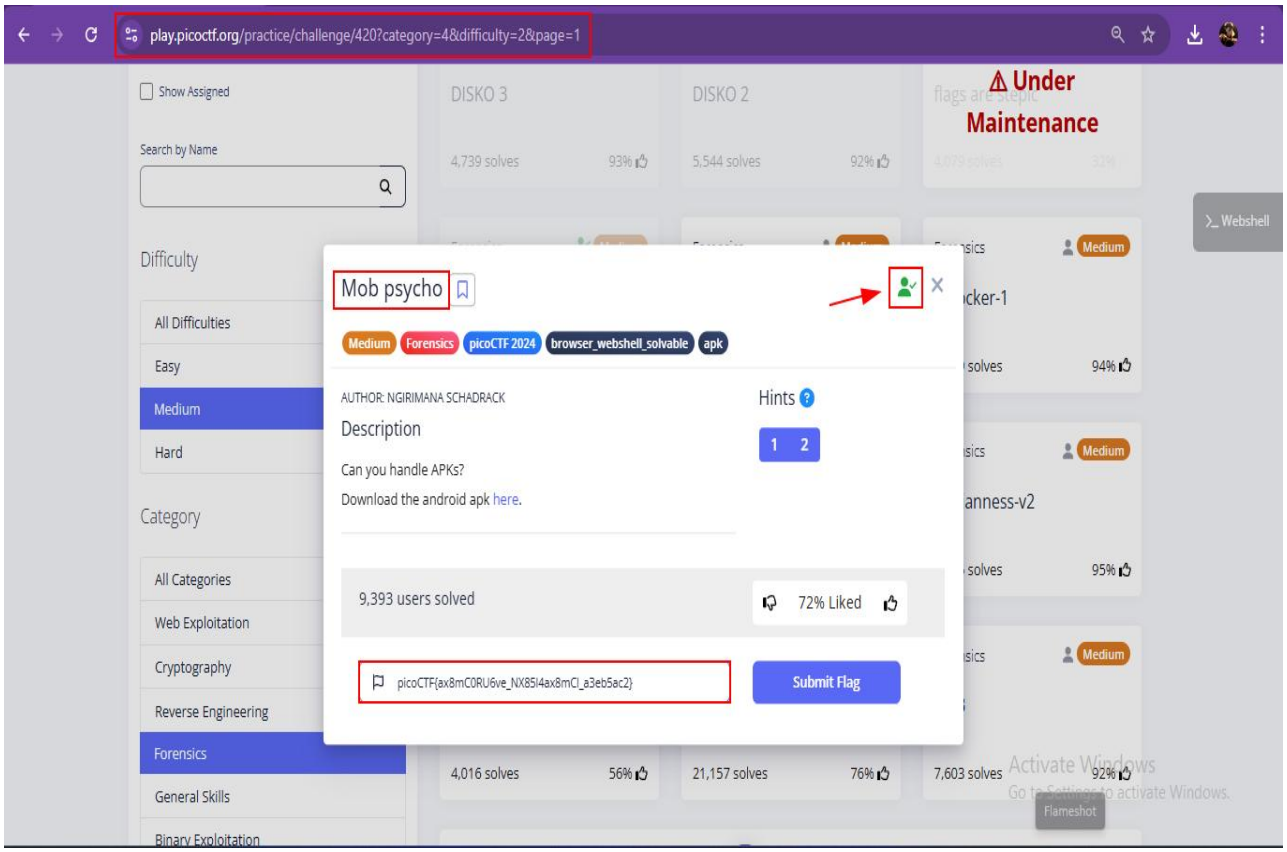
Task 3: Event-Viewing

Summary:
This challenge focused on analysing event data and improved understanding of log-based forensic investigation.



Task 4: Mob psycho

Summary:
This challenge required reviewing forensic artefacts to identify relevant evidence. It strengthened observation and analytical investigation ability.

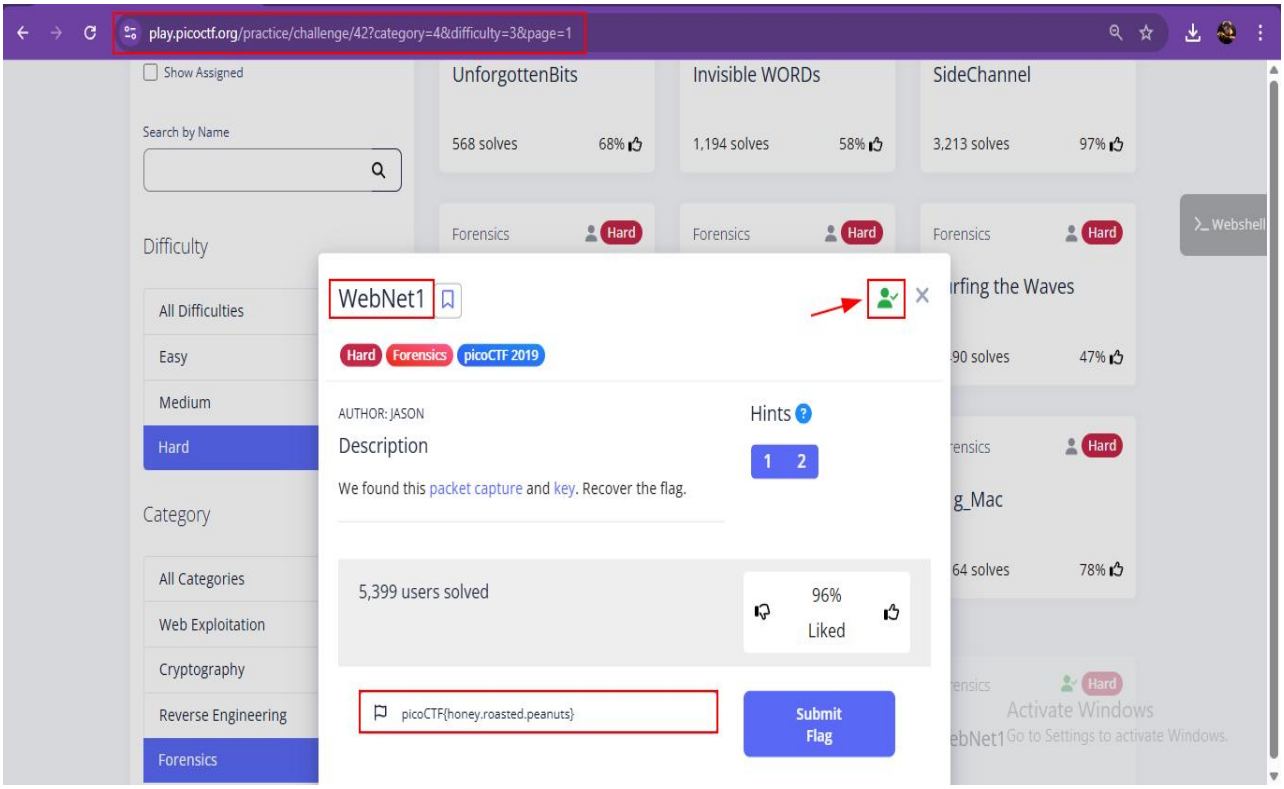


Hard Challenges

Task 1: WebNet1

Summary:

This hard challenge required advanced forensic analysis of captured data and strengthened practical investigation skills in a network-forensics context.



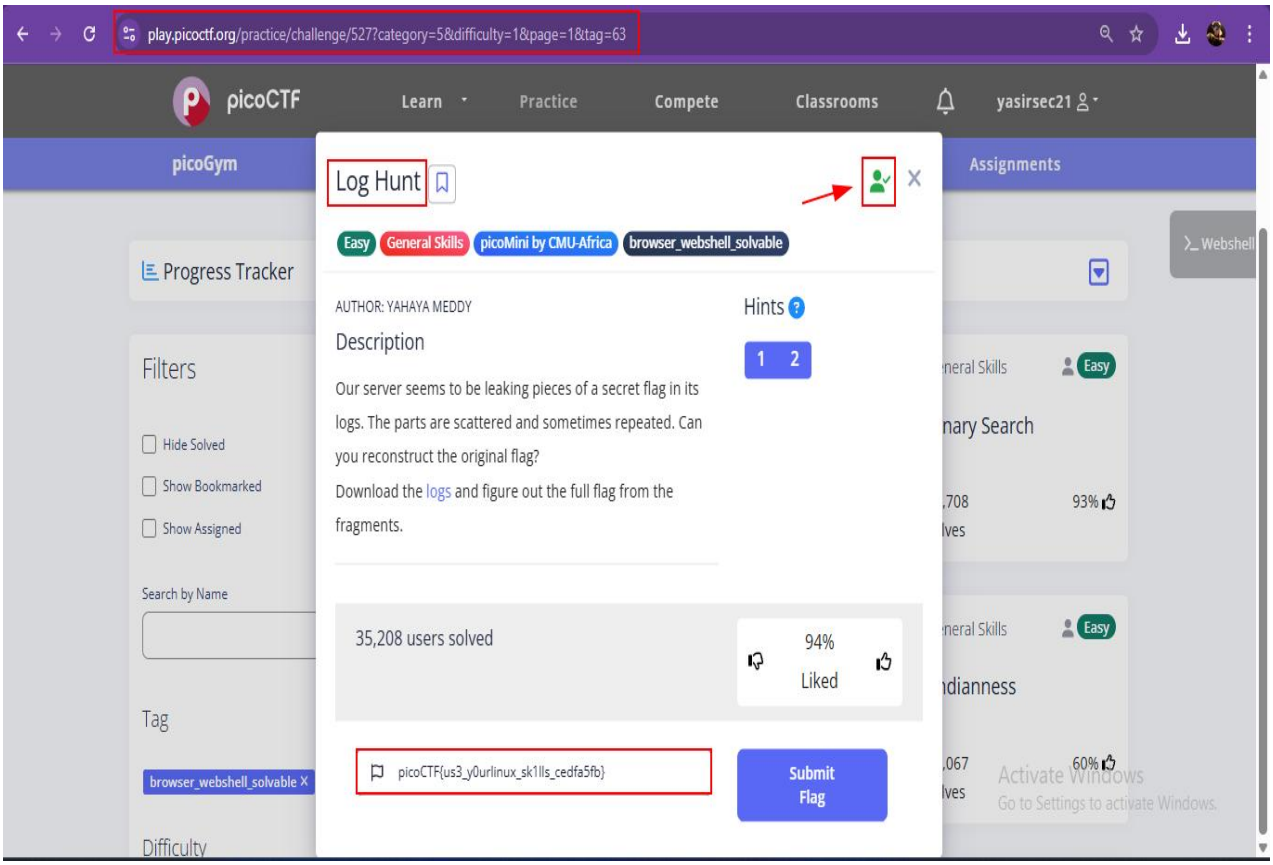
4 General Skills Challenges

Easy Challenges

Task 1: Log Hunt

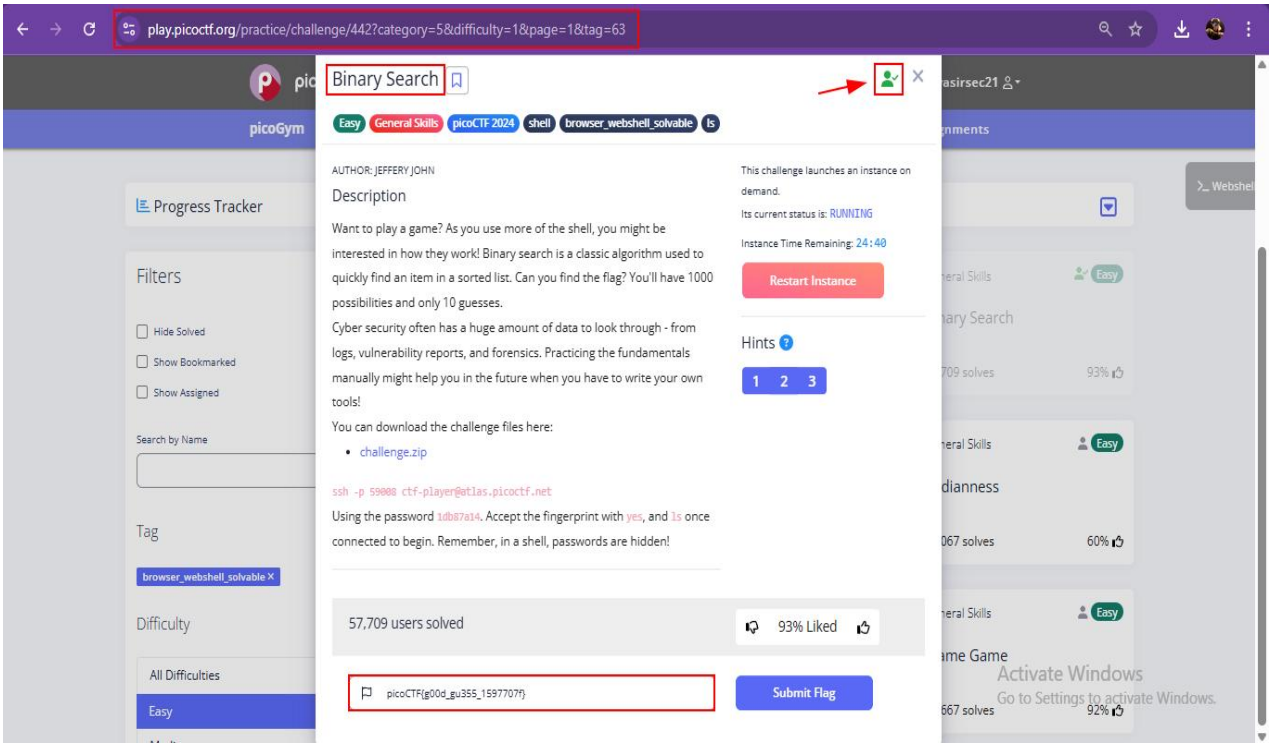
Summary:

This challenge focused on reviewing logs to identify useful information. It improved practical command-line and observation skills.



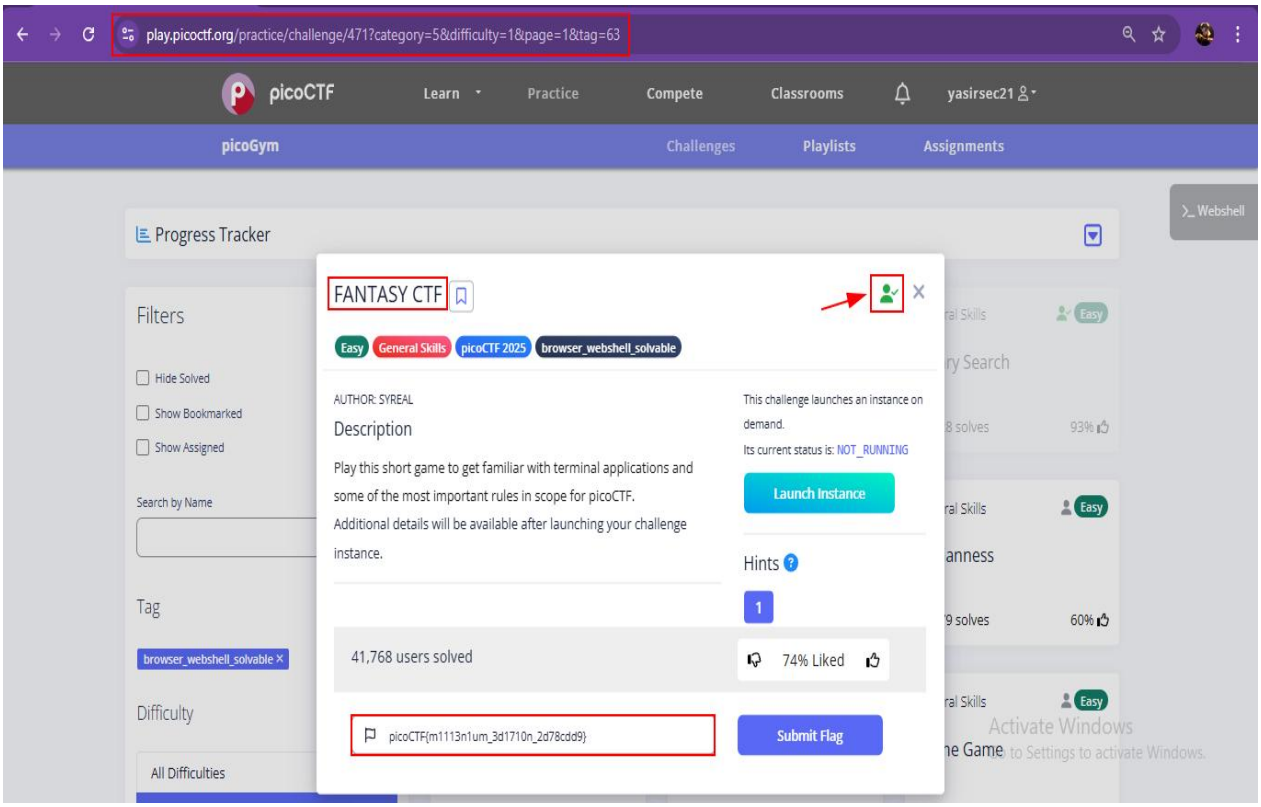
Task 2: Binary Search

Summary:
This challenge introduced structured search logic and strengthened analytical problem-solving ability.



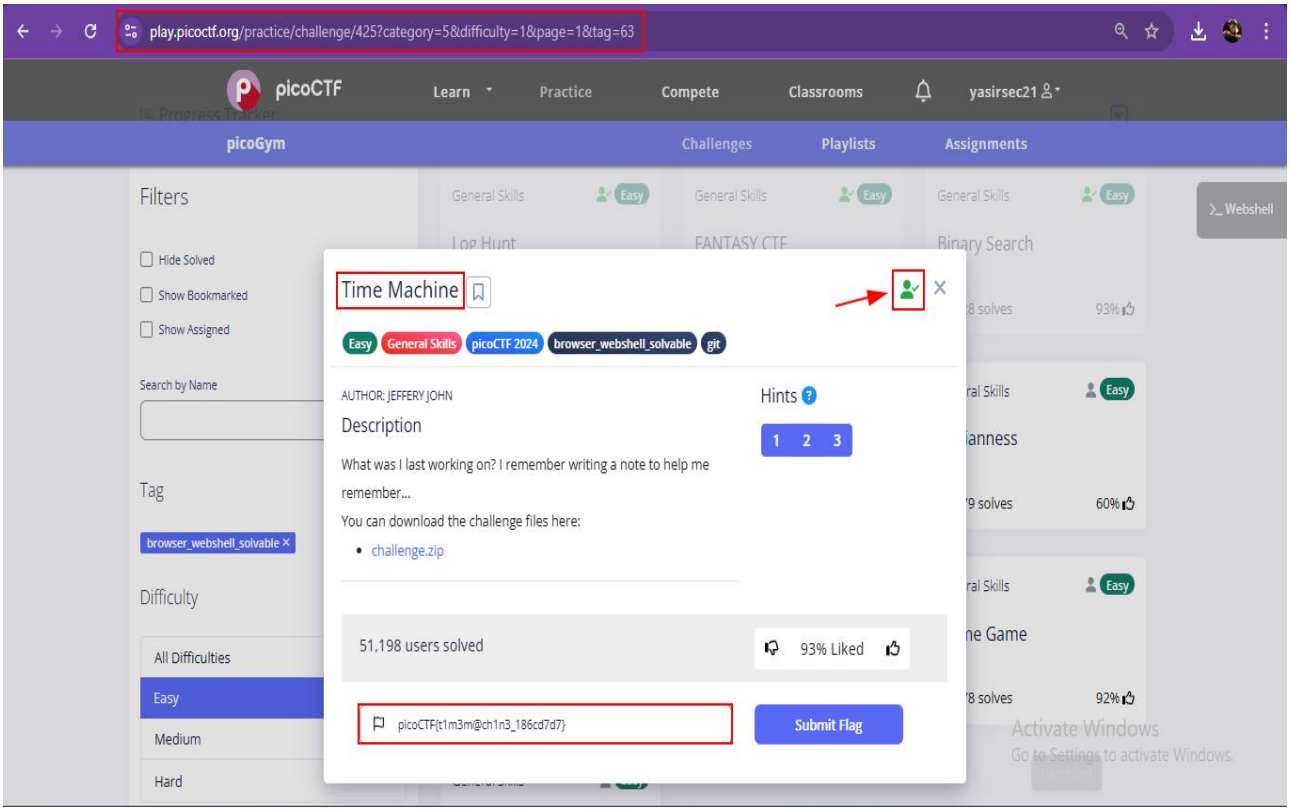
Task 3: FANTASY CTF

Summary:
This challenge was completed successfully under the General Skills category. It improved basic technical reasoning and challenge-solving confidence.



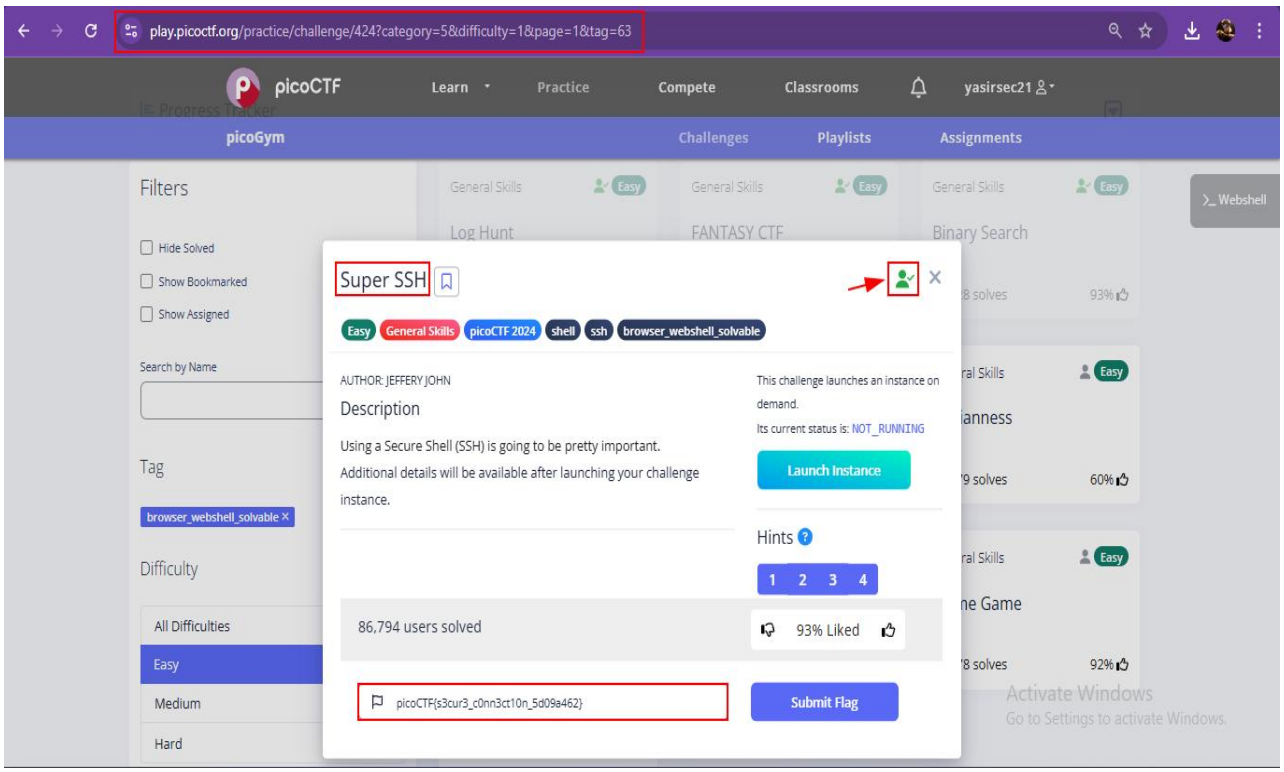
Task 4: Time Machine

Summary:
This challenge focused on recovering useful information from prior states or history.
It strengthened practical investigation and analysis skills.



Task 5: Super SSH

Summary:
This challenge improved understanding of basic remote-access concepts and strengthened command-line interaction skills.

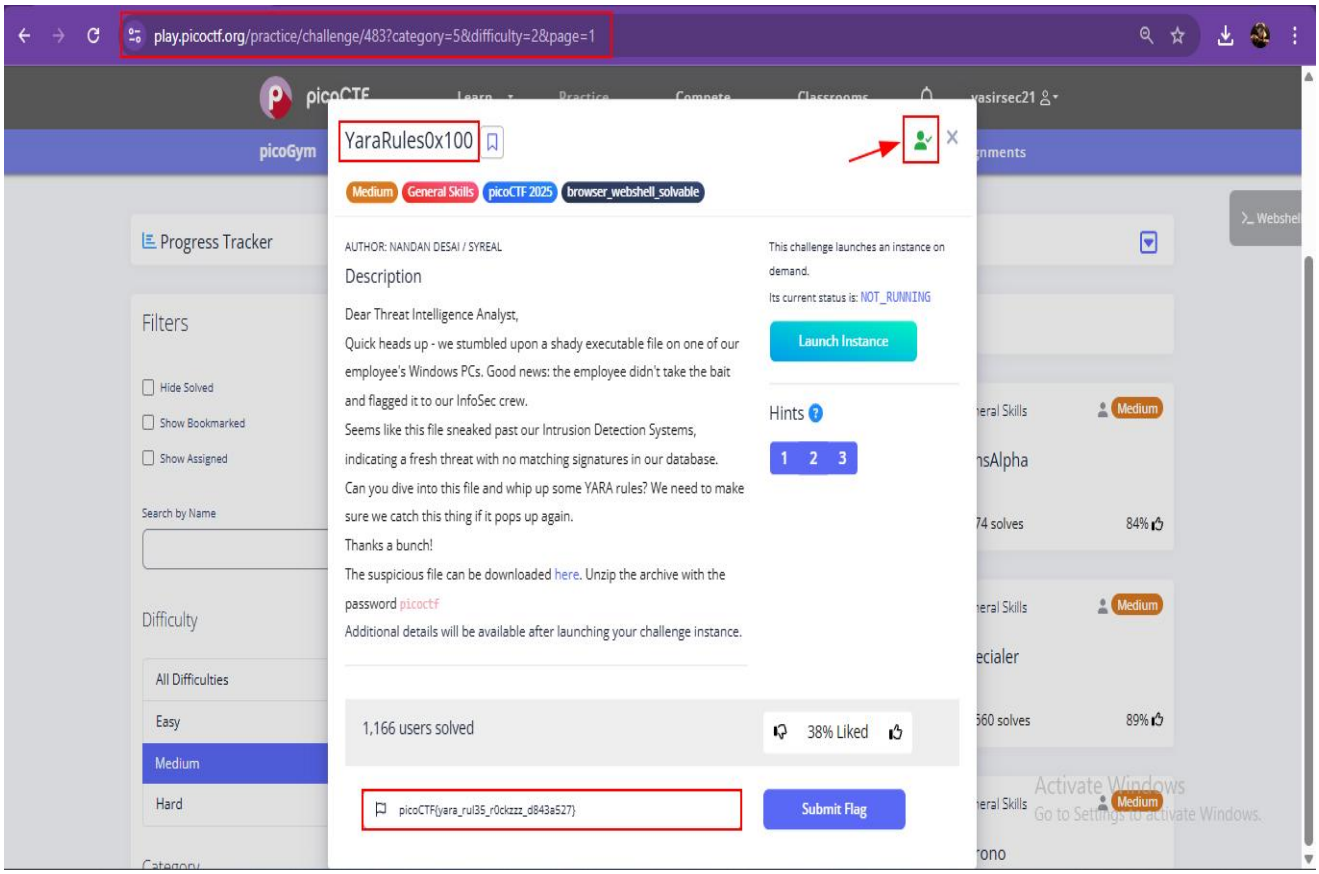


Medium Challenges

Task 1: YaraRules0x100

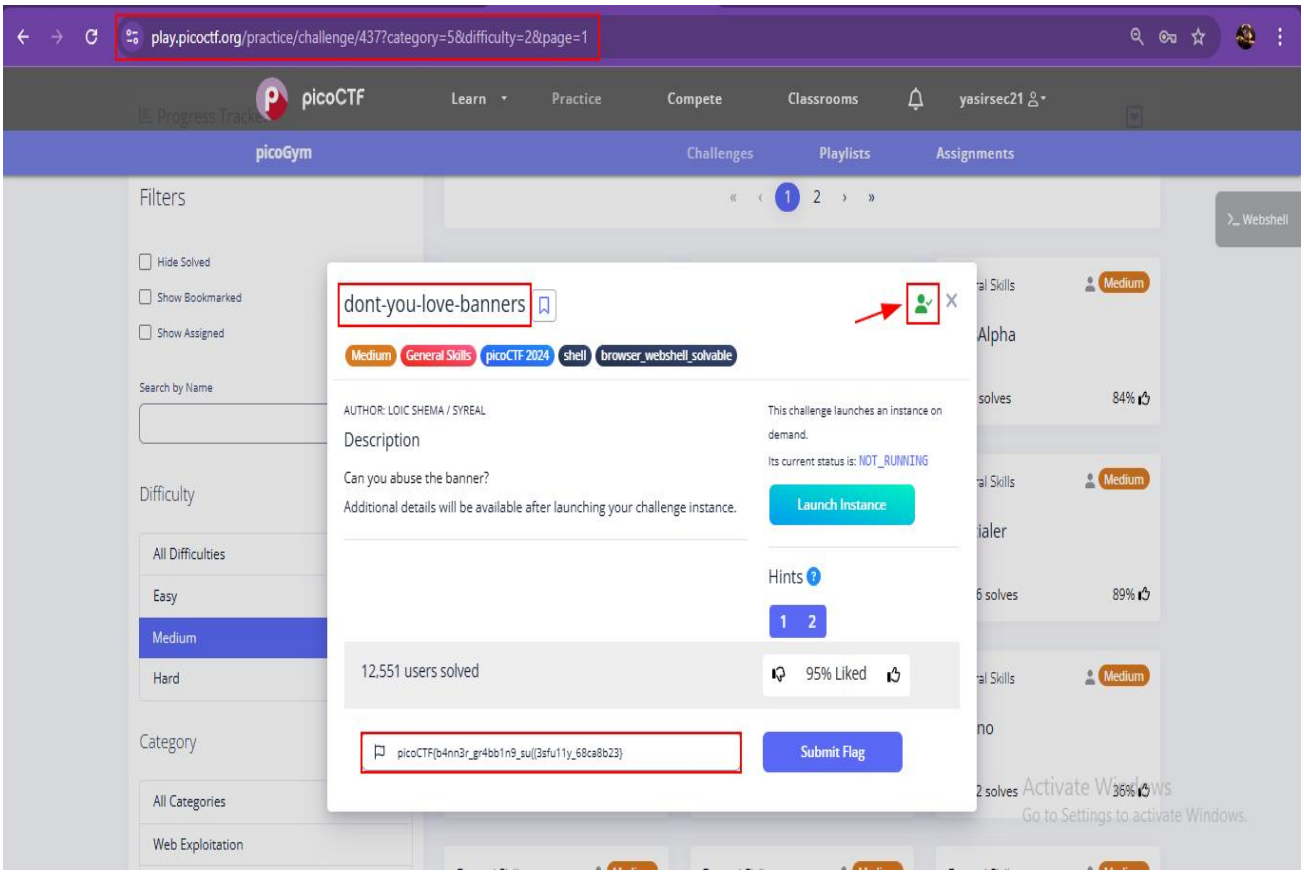
Summary:

This challenge focused on pattern-matching concepts and improved practical understanding of rule-based analysis.



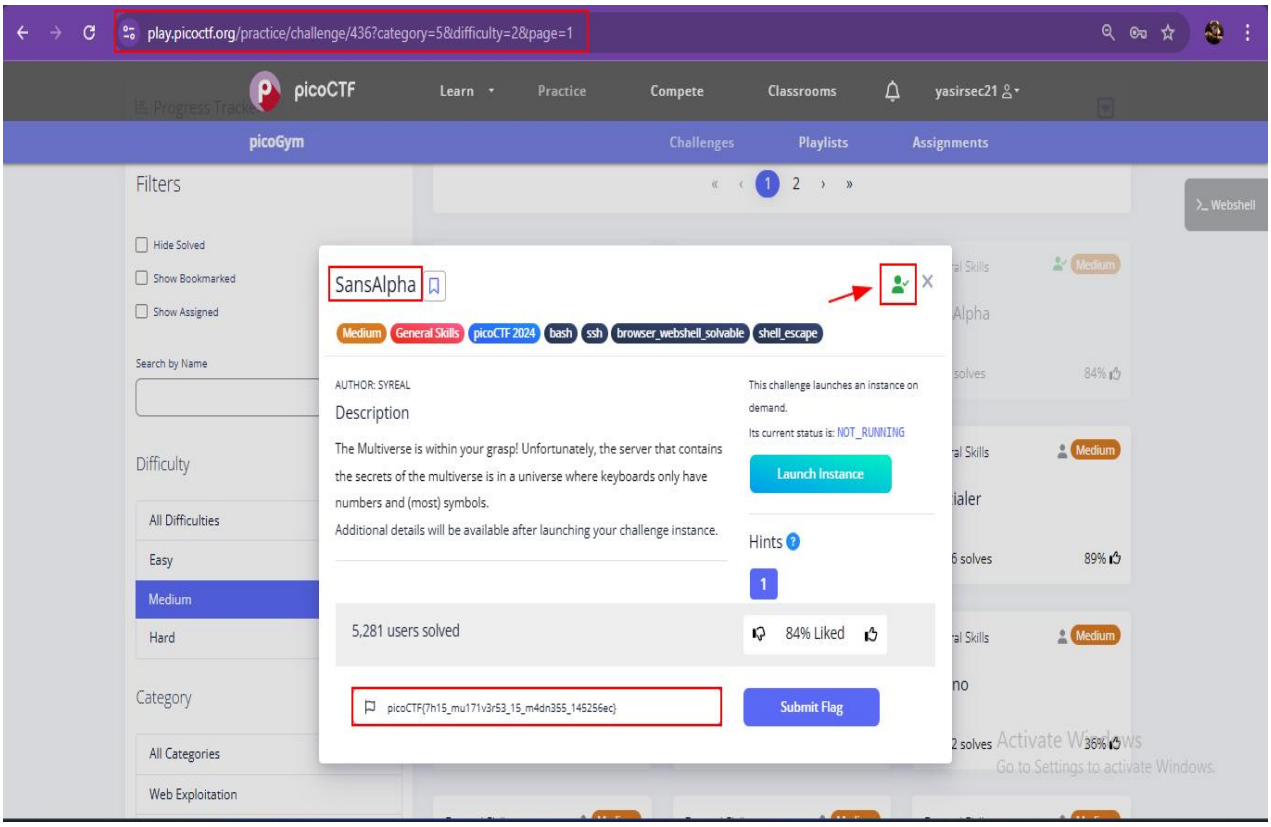
Task 2: dont-you-love-banners

Summary:
This challenge involved analysing exposed service information and strengthened observation and enumeration skills.



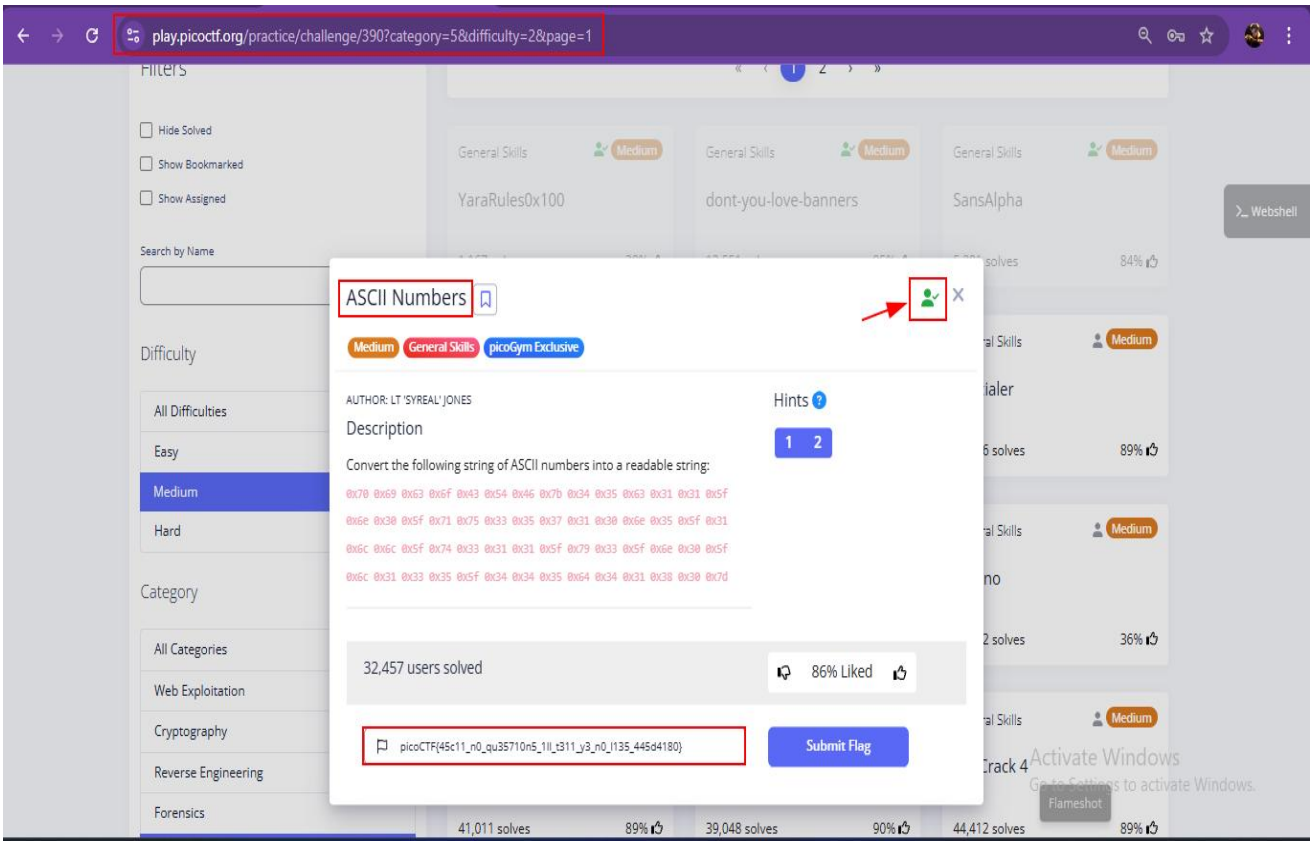
Task 3: SansAlpha

Summary:
This challenge focused on text manipulation and pattern interpretation. It improved logical reasoning and technical analysis skills.



Task 4: ASCII Numbers

Summary:
This challenge involved character encoding conversion and improved practical data interpretation skills.



PicoCTF Completion Summary

Summary:
A total of 40 PicoCTF challenges were completed successfully across Web Exploitation, Cryptography, Forensics, and General Skills categories. These challenges covered Easy, Medium, and Hard difficulty levels and provided practical exposure to real cybersecurity concepts through hands-on problem solving.

Category Breakdown

- Web Exploitation: 11 Challenges
- Cryptography: 10 Challenges
- Forensics: 10 Challenges
- General Skills: 9 Challenges

Total Completed: 40 Challenges

Conclusion

Summary:
The 40 completed PicoCTF challenges helped improve practical cybersecurity skills across multiple domains. These tasks strengthened technical understanding, analytical thinking, and hands-on problem-solving ability in a structured learning environment.

Learning Outcome

Summary:
This challenge-solving experience improved practical knowledge in Web Exploitation, Cryptography, Forensics, and General Skills, while also increasing confidence in handling cybersecurity-related tasks.

Report Submitted By

Md Yasir
Cyber Security Intern
Email: yasirsec21@gmail.com